

Mainstar®: HSM Reporter/Manager™

Tutorial

A Day in the Life of a Storage Administrator Using HSM Reporter/Manager

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Mainstar Software Corporation
P.O. Box 4132
Bellevue, WA 09009 USA

Tel: 1-425-455-3589
Fax: 1-425-455-1992
Mainstar Technical Support

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A Day in the Life of a Storage Administrator Using Mainstar®: HSM Reporter/Manager™

It's 0800, the coffee's hot; the start of another day ... and you ask yourself - Where do I begin?

1 Getting Started

May we suggest that you start the day by checking DFSMShsm's (HSM) activity from the last 24 hours using HSM Reporter/Manager? Let us show you how.

1.1 Check The HSM Log File

Two physical sequential data sets, known as HSMLOGX and HSMLOGY, together make up the HSM log. HSM records information in the LOGX data set until the LOGX data set is full. Then HSM swaps the LOGX data set with the LOGY data set, exchanges the names, and optionally informs the operator that HSM has swapped the log data sets.

When the log is switched, a message is written to the SYSLOG that you can use to trigger a batch job or an automated activity using your automated operations software, like IBM's Netview.

An example of the message written in the SYSLOG is show below.

```
ARC0020I************************************************************
ARC0027I HSM LOG SWITCHED
ARC0020I************************************************************
```

Figure 1: HSM SYSLOG 'switched' message

We recommend that our customers make the log data sets large enough to be able to control when it switches, rather than having it switch when it gets full. You can run a batch job at the desired time to force HSM to switch the logs. Use the TSO IKJEFT01 program to issue HSM's SWAPLOG command.

```
000001 //SSGCGHSM JOB (11111),'STORAGE ADMIN',
000002 //   MSGCLASS=X,CLASS=F,NOTIFY=SUBMITR,USER=SSGCG
000005 //*
000006 //STEP1   EXEC PGM=IKJEFT01
000010 //SYSPRINT DD  SYSOUT=
000011 //SYSTSPRT DD  DUMMY
000013 //SYSTSIN  DD  *
000014  HSEND SWAPLOG
```

Figure 2: Use IKJEFT01 program to force HSM SWAPLOG command

Here are some tips:

- You will want to schedule this batch job to run once every 24 hours, after all the DFSMShsm automated tasks are completed.
- Use the SETSYS ACTLOGMSGLVL(FULL) parameter in the DFSMShsm PARMLIB to ensure that DFSMShsm is logging all messages.
- Executions of DFSMShsm on other LPARS do not share a common log data set, so you will need to repeat this process for all executions of DFSMShsm in your environment.
- Check to be sure the DFSMShsm start up procedure(s) (usually found in SYS1.PROCLIB) does not swap the log when starting up (LOGSW=NO).

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1.2 Combine The Logs

In a shared environment, the HSM log data sets from each image can be combined into one output data set for HSM Reporter/Manager to report from. This batch job can be run as soon as the log files have been switched on each shared image. If you have automated operations software, you can use it to automatically trigger this batch job to run.

You will need to modify this JCL for use at your site:

- Change the JOBCARD to your installation's requirements.
- The STEPLIB points to your HSM Reporter/Manager LOAD library.
- Concatenate the HSMLOGY data set names for all images in the LOGIN DD statement. Our example is setup to include LOGY data sets from three images in a shared environment.
- The SYSLIST DD statement writes output to your specified data set or you can use an asterisk.
- The LOGOUT DD statement is the data set that is created by this process.
- The submitter requires RACF ALTER authority to create the output data set.
- We used a GDS data set with a limit of 10.
- The SYSIN DD statement points to HSM Reporter/Manager's password data set to validate the authorization code.

We are in ISPF Edit in a newly created PDS member called HSMJOB:

```
000015 //SSGCGHSM JOB (11111),'STORAGE ADMIN',
000016 //   MSGCLASS=X,CLASS=F,NOTIFY=SUBMITR
000017 //STEP1   EXEC PGM=HMPFMT
000018 //STEPLIB  DD  DSN=HSM.REPORTER.MANAGER.LOAD,DISP=SHR
000019 //SYSUDUMP DD  SYSOUT=(*,,DUMP)
000020 //LOGIN    DD  DSN=YOUR.SYSTEM.HSMLOGY1,DISP=SHR
000021 //         DD  DSN=YOUR.SYSTEM.HSM.HSMLOGY2,DISP=SHR
000021 //         DD  DSN=YOUR.SYSTEM.HSM.HSMLOGY3,DISP=SHR
000022 //SYSPRINT DD  SYSOUT=*,DCB=BLKSIZE=4096
000023 //SYSLIST  DD  DSN=YOUR.USERID.SYSLIST,DISP=SHR
000024 //HROLOG   DD  SYSOUT=*,
000025 //         DCB=(RECFM=VB,LRECL=256,BLKSIZE=7200)
000026 //LOGOUT   DD  DSN=YOUR.HLQ.EXTRACT(+1),
000027 //             DISP=(NEW,CATLG,DELETE),
000028 //             UNIT=SYSDA,
000029 //             SPACE=(TRK,(10,10),RLSE),
000030 //             DCB=(RECFM=VB,LRECL=2052,BLKSIZE=23476)
000031 //SYSIN   DD  DSN=HSM.REPORTER.MANAGER.PASSWORD(HROAUTH),
000032 //        DISP=SHR
```

Figure 3: Run Batch Job To Combine HSM Log Data Sets From Each Image And Format Out To A GDG Extract.

2 Are You New to HSM Reporter/Manager?

You will need to do some configuration tasks in order to use HSM Reporter/Manager's panels. These tasks may have been completed by the installer (check with the installer first). If they have not, you will have to complete the following tasks. Each tasks is explained in detail in the following topics.

- Establish System JCL Parameters
- Configure System Data sets
- Set System Security Parms

To complete the configuration tasks you will need to know:

- How to access HSM Reporter/Manager, either from a site-customized ISPF menu, or by executing HRPSTART from the product's CLIST library
- The names of the installation data sets
- The product authorization codes
- The names of HSM control data sets (MCDS, OCDS, BCDS)
- The system name(s) of your LPAR(s)
- The HSM LOGY data set names

2.1 Establish System JCL Parameters

Access HSM Reporter/Manager. You might see a message letting you know that the system installation is not complete. Ignore this message by pressing **Enter**.

The following is a display of the *HSM Manager Master Panel* looks like. We will refer back to the *HSM Manager Master Panel* often, so take time to familiarize yourself with it.

```
<V6 R1.xx >     HSM MANAGER MASTER PANEL
 COMMAND==>
 SUPPORT ACTIVITIES                    REPORTS FROM THE CONTROL DATA SETS

  D  DISPLAY LAST MASTER                 C1  MCDS INFORMATION
  F  PROGRAM FUNCTION KEYS               C2  BCDS INFORMATION
  J  SYSTEM JCL PARAMETERS               C3  OCDS INFORMATION
  K  SYSTEM DATA SETS                    C4  HSM HEALTH REPORTS
  S  SYSTEM SECURITY PARMS               C5  VOLUME DUMP INFORMATION
  0  GENERATE MASTER                     G   GENERATE COMMON VARIABLES

  REPORTS FROM THE LOG FILE

  1  DAILY SPACE MANAGER              5  RECYCLE RETURN CODE SUMMARY
  2  MIGRATION/BACKUP OPTIONS         6  RETURN CODE REASON CODE TABLES
  3  RECALL/RECOVER OPTIONS           7  LOG MESSAGE SUMMARY
  4  EXTENT REDUCTION SUMMARY         8  TAPE USAGE SUMMARY

  ONLINE BATCH PROCESS

  P  SELECT ONLINE PLANS FOR BATCH    B  BATCH JCL BUILD FOR ONLINES

  ENTER  END  TO TERMINATE ONLINE FUNCTION
```

Figure 4: HSM Manager Master Panel

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Key **J** (System JCL Parameters) in the Command field of the *HSM Manager Master Panel* and press **Enter**. The *HSM Manager JCL Selection Menu* panel is displayed.

The following is an example of the *HSM Manager JCL Selection Menu* panel.

```
<V6 R1.xx>             HSM MANAGER JCL SELECTION MENU
 COMMAND ===>
  Specify the JOB card to be placed in front of the JCL:
  ===> //SSGSXT JOB (P970,01),'SSGSXT',
  ===> //   MSGCLASS=S,CLASS=S,NOTIFY=SUBMITR
  ===> //*
  ===> //*
  ===> //*
  Specify the following JCL parameters:
  UNIT   ============> SYSDA     (generic or esoteric, DASD or Tape device)
  PRIMARY SPACE =====> 300       (in tracks, required if DASD)
  SECONDARY SPACE ===> 10        (in tracks, optional)
  BLKSIZE ===========> 23476     (Extract BLKSIZE)
  SYSOUT CLASS ======> *
  HRP STEPLIB NAME:
  STEPLIB  ==========> SYS3.MAINSTAR.HRP5201.LOAD

  Authorized LOADLIB  ==========> SYS3.MAINSTAR.HRP5201.LOAD
  USER PREFIX =======> SSGSXT
  PRODUCT PREFIX ====> SYS3.MAINSTAR.HSMRPT
 Enter END command to terminate this menu
```

Figure 5: HSM Manager JCL Selection Menu

The *HSM Manager JCL Selection Menu* panel (show on the previous page) is shown with all of the fields filled in. You will need to supply the values for the fields.

The following is a description of the fields on the *HSM Manager JCL Selection Menu* panel.

Field	Description
JOBCARD	Key in a valid JOBCARD to be used whenever you use HSM Reporter/Manager to submit a batch job.
UNIT	Key in a unit name, or you can use something generic like SYSALLDA. The unit parameter is used to dynamically allocate data sets but is ignored if the data sets are SMS managed.
PRIMARY SPACE	Key the number of tracks to use for the primary allocation to format the batch extraction. Remember, if you have a large log file you will need a chunk of space.
SECONDARY SPACE	It's a good idea to provide some secondary space because the number of records varies from day to day or may grow over a period of time.
BLKSIZE	This is used by the HSMLOGY format process. The RECFM is VB so if you use half track blocking (recommended) the block size is 23476.
SYSOUT CLASS	An asterisk (*) is commonly used but you can use any valid SYSOUT class for your site.

Field	Description
STEPLIB	Key in the name of the product's LOAD library. It is usually something like 'SYS3.MAINSTAR.HRP5201.LOAD'. Check with your installer.
USER PREFIX	This can be your USERID or a multi-node name you want to use for work data sets dynamically allocated by HSM Reporter/Manager. We recommend that you pick a name that is SMS managed since the data sets are dynamically allocated and you can't provide a VOLSER. Also note that if your SMS ACS routines require you to code a STORCLAS or MGMTCLAS parameter to satisfy the allocation, you will need to make allowances for these dynamic allocations (or allocation can fail).
PRODUCT PREFIX	This is the single or multi-node prefix used for the product data sets assigned to permanent data set created by HSM Reporter/Manager. In our example, we used 'SYS3.MAINSTAR'.

After you have keyed everything in, press **Enter** and then **PF3 (End)** to exit out of this panel. You will be returned to the *HSM Reporter/Manager Master Panel*.

2.2 Configure System Data Sets

Key **K** (System Data Sets) in the 'Command' field on the *HSM Reporter/Manager Master Panel* and press **Enter**. A subsequent panel, *HSM Manager Initiate Product DSNAMES*, is displayed. Key **1** (Product DSNAMES) and press **Enter**. The *HSM Manager Data Set Name Selection Menu* panel is displayed.

The following is an example of the *HSM Manager Data Set Name Selection Menu* panel.

```
<V6 R1.xx>      HSM MANAGER DATA SET NAME SELECTION MENU
 COMMAND ===>
 Specify WHOLE Data Set Names without quotes:
  INPUT TO DAILY HRP JOB:
    DSN1 ===> SSG.DASD.HSMLOG.EXTRACT(0)
    DSN2 ===>
    DSN3 ===>
    DSN4 ===>
    DSN5 ===>
    DSN6 ===>
  HRPFMT OUTLOG DSNAME => SSG.DASD.HSMLOG.EXTRACT(+1)
  MODEL (IF NOT SMS MGD=>
  PLANS DSNAME =========> SYS3.MAINSTAR.HSMRPT.PLANS
  COMMANDS DSNAME ======> SYS3.MAINSTAR.HSMRPT.COMMANDS
  CLIST LIBRARY ========> SYS3.MAINSTAR.HSMRPT.CLIST
  PROFILE DSNAME =======> SYS3.MAINSTAR.HSMRPT.PROFILE
 Enter END command to terminate this menu
```

Figure 6: HSM Manager Data Set Name Selection Menu Panel

The *HSM Manager Data Set Name Selection Menu* panel (shown above) is shown with all of the fields filled in. You will need to supply the values for the fields.

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The following is a description of the fields on the *HSM Manager Data Set Name Selection Menu* panel.

Field	Description
DSN1 THROUGH DSN6	Key in the extract data set names (1 - 6 for broader sequence) used in the batch job to reformat and copy the log file to a GDG, including the relative generation number (0) indicating the most current version. You have an option in this panel to concatenate more than one data set as input, so you can look for records from multiple days (up to five additional days). Remember this for future use.
HRPFMT OUTLOG DSNAME	This is the same GDG as above, but it's the new data set created when HRPFMT is executed, so indicate (+1).
MODEL	If the data set is SMS managed or you use DATACLAS to eliminate the need for a model DSCB, you don't need to key anything in here. If you do need a model DSCB, key in the name here.
PLANS DSNAME	***** Key in the name of the product's plans data set name.
COMMANDS DSNAME	***** Key in the name of the product's COMMANDS data set name.
CLIST LIBRARY	***** Key in the name of the product's CLIST library.
PROFILE DSNAME	***** Key in the name of the product's PROFILE library.

***** Refer to the *HSM Reporter/Manager Installation & User Guide* (document 002-0202)

After you have keyed everything in, press **Enter** and then **PF3 (End)** to exit out of this panel. You will be back to the *HSM Manager Initiate Product DSNAMES* panel.

This time, key **2** (Control Data Sets) on the *HSM Manager Data Set Names Selection* panel and press **Enter**. The *HSM Manager Control Data Set Names Selection* panel is displayed.

The following is an example of the *HSM Manager Control Data Set Names Selection* panel.

```
<V6 R1.xx>      HSM MANAGER CONTROL DATA SET NAMES SELECTION
 COMMAND ===>
 KEY S TO ESTIMATE EXTRACTION SPACE FROM LOG FILE ==>
  MCDS DSNAME ===> HSM.MCDS
  MCDS1 DSNAME ==> NULLFILE
  MCDS2 DSNAME ==> NULLFILE
  MCDS3 DSNAME ==> NULLFILE
  BCDS DSNAME ===> HSM.BCDS
  BCDS1 DSNAME ==> NULLFILE
  BCDS2 DSNAME ==> NULLFILE
  BCDS3 DSNAME ==> NULLFILE
  OCDS DSNAME ===> HSM.OCDS
  OCDS1 DSNAME ==> NULLFILE
  OCDS2 DSNAME ==> NULLFILE
  OCDS3 DSNAME ==> NULLFILE
  ID1 => NWL1       LOGY 1==> HSM.HSMLOGY1
  ID2 => NC50       LOGY 2==> HSM.HSMLOGY2
  ID3 =>            LOGY 3==>
  ID4 =>            LOGY 4==>
  ID5 =>            LOGY 5==>
  ID6 =>            LOGY 6==>
 Enter END command to terminate this menu
```

Figure 7: HSM Manager Control Data Set Names Selection Panel

Enter the names of your HSM control data sets on the *HSM Manager Control Data Set Names Selection* panel. Some sites had to split them up because they exceeded IBM's 4GB limit. If your CDSs are all in one piece, just ignore those extra fields.

The following is a description of the fields on the *HSM Manager Control Data Set Names Selection Menu* panel.

Field	Description
MCDS DSNAME	Key the name(s) of your migration control data set(s).
BCDS DSNAME	Key the name(s) of your backup control data set(s).
OCDS DSNAME	Key in the name(s) of your offline control data set(s).
ID1 - ID6	Key in the system name of each LPAR you run HSM on.
LOGY1 - LOGY6	Key in the name of each HSM LOGY data set (one for each LPAR).

When you have entered your file names, press **Enter** and then **PF3** to return to the *HSM Manager Initiate Product DSNAMES* panel.

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Key **3** (Specify HSM Release Number) on the *HSM Manager Initiate Product DSNAMES* panel, and press **Enter**. The *HSM Manager HSM Release Panel* is displayed.

The following is an example of the *HSM Manager HSM Release Panel*.

```
<V6 R1.xx>     HSM MANAGER HSM RELEASE PANEL
 COMMAND==>

  SPECIFY HSM RELEASE NUMBER:

       HSM RELEASE  ========> HSM1.1  HSM1.1 (OR ABOVE)
                                      DFHSM2.6    (OR BELOW




 ENTER END COMMAND TO TERMINATE THIS MENU
```

Figure 8: HSM Manager HSM Release Panel

This panel is used to differentiate DFHSM users from HSM users.

In the HSM release field:

- If you are running DFSMShsm - key 'DFHSM2.6'
- If you are running DFSMShsm 1.1 (or above) - key 'HSM1.1'.
  Note: if you are running above 1.1, do not put in the actual release number.

Press **Enter** and then **PF3** to exit out of this panel. The *HSM Manager Initiate Product DSNAMES* panel is displayed.

You are done *HSM Manager Initiate Product DSNAMES* panel, so press **PF3** again. The *HSM Manager Master Panel* is displayed.

2.3 Set System Security Parms

The final configuration item is to point HSM Reporter/Manager at the security authorization member so we can validate the code. Key **S** in the 'Command' field of the *HSM Manager Master Panel* and press **Enter**. The *HSM Manager Security Options* panel is displayed (not shown). Key **S** in the 'Command' field and press **Enter**.

The *HSM Reporter/Manager Security Parameters* panel is displayed. Key in the name of the HSM Reporter/Manager data set that contains the member 'HROAUTH'. This member is normally found in the 'PROFILE' data set. The person who installed HSM Reporter/Manager on your system should have the security codes already setup for you.

```
<V6 R1.xx>             HSM Reporter/Manager Security Parameters
 COMMAND ===>

  Specify the data set containing security parameters for HSM Reporter/Manager.
  If the security DSN is blank, enter the security name (no apostrophizes).
  A partitioned data set will be allocated and the security member will
  be primed with a security default pattern.

  Specify the data set containing security parameters for HSM Reporter/Manager.

  Security DSN===> SYS3.MAINSTAR.HSMRPT.PASSWORD

 Hit ENTER to display/modify security member
 Enter END command to terminate this menu
```

Figure 9: Example of a Security Parameters panel with the PROFILE Data Set Filled In

Press **Enter** to display an ISPF Edit panel for your PROFILE data set.

```
EDIT       SYS3.MAINSTAR.HSMRPT.PASSWORD(HROAUTH) - 01.00    Columns 00001 00072
 Command ===>                                                  Scroll ===> PAGE
 ****** ***************************** Top of Data ******************************
 000001 AUTHORIZATION( + /* SECURITY START */
 000002  AUTHCODE(7E47ADF7C4C1) +
 000003  JULDATE(001581) +
 000004  CPUID01(01C0502030115200) +
 000005  CPUID02(F021000040716000) +
 000006  DSNAME(&&TEMP) +
 000007  UNIT(SYSALLDA) +
 000008  )
 ****** **************************** Bottom of Data ****************************
```

Figure 10: Example of Authorization Codes in PROFILE Data Set

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3 Look at the Data Using HSM Reporter/Manager

Access HSM Reporter/Manager and key **O** (GENERATE MASTER) in the 'Command' field of the *HSM Manager Master Panel* and press **Enter**. The *HSM Manager Select Log File To Summarize* panel is displayed.

The following is an example of the *HSM Manager Select Log File To Summarize* panel.

```
<V6 R1.xx>       HSM MANAGER SELECT LOG FILE TO SUMMARIZE
 COMMAND ===>
  SELECT DATA SET FOR ANALYSIS:
   SELECT     SYSID    DATA SET NAME
   ------     -------- -------------------------------------
        KEY=>   S TO RECOMPUTE EXTRACTON SPACE ESTIMATES
        ID1=> NWL1     LOGY 1==> HSM.HSMLOGY1
        ID2=> NC50     LOGY 2==> HSM.HSMLOGY2
        ID3=>          LOGY 3==>
        ID4=>          LOGY 4==>
        ID5=>          LOGY 5==>
        ID6=>          LOGY 6==>
   S     FORMATTED LOG==> SSG.DASD.HSMLOG.EXTRACT(0)
         OTHER LOG FILE=>

   MSG SPACE==> 300     MIG SPACE== 500     FORMAT SPACE==> 300

  ENTER S OR BLANK UNDER SELECT TO CHOOSE TYPE OF DATA SET

  SPECIFY FULLY QUALIFIED DATA SET NAME (WITHOUT QUOTES) AFTER ARROW
 ENTER END COMMAND TO TERMINATE THIS MENU
```

Figure 11: HSM Manager Select Log File To Summarize Panel

The 'GENERATE MASTER' option processes raw HSM LOGY data in a format that can be effectively used by HSM Reporter/Manager to see HSM activity and errors, or reformats the extract file for online usage. You need to complete the 'GENERATE MASTER' option, either interactively or in batch each time a swap log is done.

The ID fields should reflect the name(s) of your system(s) and the name(s) of the LOGY data set(s) for the system(s). The *HSM Manager Select Log File To Summarize* panel (show above) shows system 'NWL1', with LOGY data set name 'HSM.HSMLOGY1' and system 'NC50' with LOGY data set 'HSM.HSMLOGY2'. If the fields are not filled in, you missed something in the configuration steps above. Review that section before continuing.

The 'FORMATTED LOG' field contains the name of the output GDG data set created by our batch job (see JCL). We have indicated that we want to format the most current generation (0) of the GDG.

Further down on the *HSM Manager Select Log File To Summarize* panel there are three fields to indicate space needs. The amount of space required differs for everybody and is based on how much HSM activity is done within the 24-hour period (and how much data is in the log data sets). You may need to experiment with the space requirements until you have enough space to format the log. Select the 'FORMATTED LOG' by keying **S** next to the field on the panel and press **Enter** to start the process.

You will see a little pop-up panel that provides the status of the event.

```
<V6 R1.xx>       HSM MANAGER SELECT LOG FILE TO SUMMARIZE
 COMMAND ===>
  SELECT DATA SET FOR ANALYSIS:
   SELECT     SYSID    DATA SET NAME
   ------     -------- -------------------------------------
        KEY=>   S TO RECOMPUTE EXTRACTION SPACE ESTIMATES
        ID1=> NWL1   .-----------------------------------.
        ID2=> NC50   | HSM REPORTER STATUS PANEL          |
        ID3=>        |      STATUS                        |
        ID4=>        |                                    |
        ID5=>        | ALLOCATING WORKFILES               |
        ID6=>        |                                    |
   S     FORMATTED L | 10:40:13                           |
         OTHER LOG F |                                    |
                     |                                    |
   MSG SPACE==> 300  |                                    |300
                     |                                    |
  ENTER S OR BLANK U .-----------------------------------.

  SPECIFY FULLY QUALIFIED DATA SET NAME (WITHOUT QUOTES) AFTER ARROW
 ENTER END COMMAND TO TERMINATE THIS MENU
```

Figure 12: HSM Reporter Status Pop-Up Panel For Generate Master Task

When the formatting is complete, you will see a *HSM Manager Logfile Summary Panel*. The information on this panel is extracted from the logfile. This panel provides the current date and time so you can compare it to the start and end time of the log data. The total elapsed time is how many hours of data collected in this version of the logfile. The last line is the total record count in the log.

An example of the *HSM Manager Logfile Summary Panel* is show below.

```
COMMAND===>

 <V6 R1.xx>  HSM MANAGER LOGFILE SUMMARY PANEL
                    FOR SSG.DASD.HSMLOG.EXTRACT(0)

   CURRENT DATE AND TIME========>YYYY/DDD   HH:MM:SS

   LOGFILE START DATE AND TIME==>YYYY/DDD   HH:MM:SS
   LOGFILE END DATE AND TIME====>YYYY/DDD   HH:MM:SS
   LOGFILE ELAPSED TIME=========>0006:25:33

   RECORD COUNT ON LOGFILE======>132982


  ENTER  END  TO CONTINUE WITH ONLINES
```

Figure 13: HSM Manager Logfile Summary Panel For The Selected Formatted Log

Press **PF3** to exit out of this panel. *The HSM Manager Master Panel* is again displayed.

> **Note:** The 'GENERATE MASTER' option processes raw HSM LOGY data and creates an output file that is used by HSM Reporter/Manager to display HSM activity and errors. You will need to complete the 'GENERATE MASTER' option daily to process each day's HSM LOGY data.

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3.1 Reports From the Log File - Log Activity

On the left hand side of the *HSM Manager Master Panel* is the 'REPORTS FROM THE LOG FILE' section.

Options 1 - 4 reflect the data in the log file we formatted with the 'GENERATE MASTER' option. This is where we look at HSM's activity over the last 24 hours.

The following is an example of the *HSM Manager Master Panel*.

```
<V6 R1.xx>     HSM MANAGER MASTER PANEL
 COMMAND==>
 SUPPORT ACTIVITIES                    REPORTS FROM THE CONTROL DATA SETS

  D  DISPLAY LAST MASTER                 C1  MCDS INFORMATION
  F  PROGRAM FUNCTION KEYS               C2  BCDS INFORMATION
  J  SYSTEM JCL PARAMETERS               C3  OCDS INFORMATION
  K  SYSTEM DATA SETS                    C4  HSM HEALTH REPORTS
  S  SYSTEM SECURITY PARMS               C5  VOLUME DUMP INFORMATION
  0  GENERATE MASTER                     G   GENERATE COMMON VARIABLES

  REPORTS FROM THE LOG FILE

  1  DAILY SPACE MANAGER              5  RECYCLE RETURN CODE SUMMARY
  2  MIGRATION/BACKUP OPTIONS         6  RETURN CODE REASON CODE TABLES
  3  RECALL/RECOVER OPTIONS           7  LOG MESSAGE SUMMARY
  4  EXTENT REDUCTION SUMMARY         8  TAPE USAGE SUMMARY

  ONLINE BATCH PROCESS

  P  SELECT ONLINE PLANS FOR BATCH    B  BATCH JCL BUILD FOR ONLINES

  ENTER  END  TO TERMINATE ONLINE FUNCTION
```

Figure 14: REPORTS FROM THE LOG FILE Section Of The HSM Manager Master Panel

3.2 Daily Space Manager - Master Panel Option 1

This option provides a summary of activity generated for each DASD volume that HSM performed automatic space management functions. Key **1** in the 'Command' field of the *HSM Reporter Master Panel* and press **Enter**. The *HSM Manager Daily Space Manager* panel is displayed.

The following is an example of the *HSM Manager Daily Space Manager* panel. In this example, a few of the 44 total DASD volumes are displayed.

```
<V6 R1.xx> HSM MANAGER DAILY SPACE MANAGER                Row 1 to 18 of 44
 OPTION ===>                                                 SCROLL==> PAGE
  SORT BY==> VOLSER   VOLSER, COUNT, HPCF, LPCF, HCYF, LCYF, HFRG, LFRG
  SORT TYPE> A        A FOR ASCENDING, D FOR DESCENDING
             PROC  HI    LOW    HI     LOW     HIFRG LOFRG
 S  A VOLSER COUNT FREE  FREE   CYLS   CYLS    INDEX INDEX
      HSMM01 00001  27    27    00802  00802   .440  .440
      HSMM02 00001  33    33    01035  01035   .398  .398
      HSMM03 00001  31    31    00959  00959   .362  .362
      HSMM04 00001  22    22    00618  00618   .477  .477
      HSMM05 00001  31    31    00968  00968   .346  .346
      HSMM06 00001  36    36    01146  01146   .293  .293
      HSMM07 00001  52    52    01693  01693   .281  .281
      HSMM08 00001  37    37    01152  01152   .273  .273
      HSMM09 00001  40    40    01251  01251   .301  .301
      HSMM10 00001  27    27    00865  00865   .322  .322
      HSMM11 00001  29    29    00897  00897   .425  .425
      HSMM12 00001  32    32    00989  00989   .399  .399
      HSMM13 00001  35    35    01097  01097   .337  .337
      HSMM17 00001  33    33    01037  01037   .366  .366
      HSMM18 00001  29    29    00907  00907   .275  .275
```

Figure 15: HSM Manager Daily Space Manager Panel

The following is a description of the fields in the *HSM Manager Daily Space Manager* panel.

Field	Description
VOLSER	The volume serial number of each DASD volume.
PROC COUNT	The number of times HSM space management functions provided a service for this VOLSER within the reporting interval.
HI FREE	The highest percent of free space found on this volume within the interval.
LOW FREE	The lowest percent of free space found on this volume.
HI CYLS	The highest number of cylinders available for allocation on this volume.
LOW CYLS	The lowest number of cylinders available for allocation on this volume.
HIFRG INDEX	The highest fragmentation index found for this volume.
LOFRG INDEX	The lowest fragmentation index found for this volume.

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In addition to the fields that display data, there are options available for you to use to manipulate the display. The following is a description of these options.

Option	Description
SORT Option **Sort By** **Sort Type**	To sort this display based on your criteria: Key 'SORT' in the 'OPTION' field. Key one of the field names (VOLSER, COUNT, HPCF (high percent free), LPCF (low percent free), HCYF, LCYF, HFRG, LFRG) in the 'SORT BY' field. Key **A** for ascending or **D** for descending in the 'Sort Type' field. Press **Enter.**
P Option (print display)	Key **P** in the 'OPTION' field to print the data display. All of the information is printed, not just the currently visible page. A batch job is submitted to generate the printed output.
S (select)	Key **S** in the 'Select' column next to any of the DASD volumes displayed and press **Enter**. The *HSM Reporter Daily Space Manager Selected Detail* panel is displayed. You may select more than one DASD volume at a time. The displayed panel includes all of the start, end, and elapsed times; data sets and tracks freed as a result of migration; minimum age of migrated data sets, free tracks, free cylinders, and fragmentation indexes. Press **PF3** to end out of the *HSM Reporter Daily Space Manager Selected Detail* panel.
P (print selected volumes)	Key **P** in the 'Select' column next to any of the DASD volumes displayed and press **Enter** to print a hard copy of the selected detail for that DASD volume. You may key **P** next to more than one DASD volume at a time. A batch job is submitted to generate the printed output.
A (action) column	The 'Action' column on this panel displays the last action that you keyed in for each DASD volume displayed. For example, if you key **P** and press **Enter**, the 'Action' column displays the letter **P** when you return to the *HSM Manager Daily Space Manager* panel.
B (backup)	Key **B** in the 'Select' column and press **Enter** to look at backup actions performed for data sets on the DASD volume.
M (migration)	Key **M** in the 'Select' column and press **Enter** to look at migration actions performed for data sets on the DASD volume.
R (recall)	Key **R** in the 'Select' column and press **Enter** to look at recall actions performed to this DASD volume.

Option	Description
V (recover)	Key **V** in the 'Select' column and press **Enter** to look at recover actions performed to this DASD volume.
E (extent reduction)	Key **E** in the 'Select' column and press **Enter** to look at the extent reduction actions performed for data sets on this DASD volume.

Wow! Lots of good information provided from this summary panel. As always, use **PF3** to exit out of the panel. You will be returned to the *HSM Manager Master Panel*.

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3.3 Migration/Backup Options - Master Panel Option 2

This is usually the first place that we go (not that the summary information isn't useful) because we need to know what got migrated over the last 24 hours, what didn't, and why. We can also fix some of the problems identified in these panels interactively and complete the failed migration or backup functions with HSM commands.

Key **2** in the 'Command' field on the *HSM Manager Master Panel* and press **Enter**. The *HSM Manager Logfile Migration/Backup Options* panel is displayed.

The following is an example of the *HSM Manager Logfile Migration/Backup Options* panel.

```
<V6 R1.xx>     HSM MANAGER LOGFILE MIGRATION/BACKUP OPTIONS

 COMMAND==>

               1  MIGRATION RETURN CODE SUMMARY

               2  ML1 TO ML2 RETURN CODE SUMMARY

               3  BACKUP RETURN CODE SUMMARY

               4  FILTER OPTIONS FOR MIGRATE/BACKUP

               5  DISPLAY MIGRATION/BACKUP PLANS

               6  DISPLAY MIGRATION/BACKUP FILTERED FLAT FILES


  ENTER  END  TO TERMINATE ONLINE FUNCTION
```

Figure 16: HSM Manager Logfile Migration/Backup Options Panel

3.3.1 Logfile Migration/Backup Option 1 - Migration Return Code Summary

Key **1** in the 'Command' field of the *HSM Manager Logfile Migration/Backup Options* panel and press **Enter** to display the migration return code summary. Unlike the summary information provided 'by DASD volume' (**Option 1** of the *HSM Reporter Master Panel*), we can look at 'all of the migration activity' (ALLVOL) in the *Migrate Summary* panel.

The *Migrate Summary* panel is going to look different for everyone because the types of errors and conditions are different in each site.

The following is an example of the *Migrate Summary* panel.

```
Row 1 of 9
 <V6 R1.xx>       HSM MANAGER MIGRATE SUMMARY
 OPTION ===>                                              SCROLL==> PAGE

                            FOR ALLVOL

 S    RC  COUNT   MESSAGE
      00  000593  MIGRATION OK
 S    17  000003  PDS DIRECTORY READ ERROR
      19  000025  DATA SET IN USE
      30  000004  DATA SET NOT CATALOGED
      37  000044  NO SPACE ON MIG VOL
      51  000001  DATA SET IN USE BY DFHSM
      58  000001  MIGRATION OR DBA DBU FAILED
      70  000015  SMS DS ERROR
      99  002451  UNSUPPORTED DS
 ****************************** Bottom of data ******************************
```

Figure 17: HSM Manager Migrate Summary Panel Displays Migration Return Codes For All Volumes

The *Migrate Summary* panel (shown above) shows that:

- **593** data sets were migrated successfully (RC 0).
- **3** data sets received a return code of **17**, indicating that a PDS directory read error occurred.
- **25** data sets were in use at the time space management ran, data sets were not cataloged.
- **44** could not migrate because there wasn't enough space available on ML1 (DASD).
- **15** have a DFSMS (SMS) error.
- **2,451** are unsupported (wow!).

We are going to examine some of these readings, starting with the three data sets with the RC17. Key **S** in the 'Select' column next to the return code 17 and press **Enter**. The *HSM Reporter Selected Migrate Return Code* panel is displayed.

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The following is an example of the *HSM Reporter Selected Migrate Return Code* panel.

```
<V6 R1.xx> HSM REPORTER SELECTED MIGRATE RETURN CODE              Row 1 of 3
 OPTION ===>                                                   SCROLL==> PAGE
  ENTER O ON AT OPTION FOR LIST OF OPTIONS.
  PANEL 1 OF 3. SCROLL RIGHT FOR MORE INFORMATION.
  KEY S BESIDE ENTRY FOR RETURN/REASON CODES
 S   DSN                                        ACTION   RC RSNC  TRKS  AGE
    GTO.REL315.ADA.NATSOUR.D284                 MIGRATE  17 00000    28   36
    AUD.TUPPER.WAREWL.CAN.TABLE                 MIGRATE  17 00000    75   36
    LMN.FTP.CLWWL4                              MIGRATE  17 00000    75    5
 ******************************* Bottom of data ********************************
```

Figure 18: HSM Reporter Selected Migrate Return Code Panel

The *HSM Reporter Selected Migrate Return Code* panel (shown above) displays the three data sets in error. This panel provides great information about the failed migration task. The first data set, **GTO.REL315.ADA.NATSOUR.D284** failed migration with a **return code of 17**, **reason code 0000**. The data set's size is **28 tracks**, and it's been **eligible for migration for 36 days**; but because of this error, it can not migrate.

If I scroll right, more information is provided. (To scroll right, press **PF11**). The *HSM Reporter Selected Migrate Return Code* panel scrolled right is displayed below.

```
<V6 R1.xx> HSM REPORTER SELECTED MIGRATE RETURN CODE              Row 1 of 3
 OPTION ===>                                                   SCROLL==> PAGE
  ENTER O ON AT OPTION FOR LIST OF OPTIONS.
  PANEL 2 OF 3. SCROLL LEFT OR RIGHT FOR MORE INFORMATION.
  KEY S BESIDE ENTRY FOR RETURN/REASON CODES
 S   DSN                                          DATE    TIME     FM VOL TO VOL
    GTO.REL315.ADA.NATSOUR.D284                 2000048 13:51:05   NWLL88 PL1005
    AUD.TUPPER.WAREWL.CAN.TABLE                 2000048 13:53:25   PXS004 PL1005
    LMN.FTP.CLWWL4                              2000048 14:00:22   PXS003 PL1012
 ******************************* Bottom of data ********************************
```

Figure 19: HSM Reporter Selected Migrate Return Code Panel - scrolled right

In this example, we can see the date that HSM attempted the function, the time, what DASD volume that data set lives on, and what volume the data set would have been migrated to if the migration function worked. A good thing to do now is to fix the PDS Directory Read Error for each of these data sets. It's important to fix the errors found by HSM Reporter/Manager because HSM can not perform any function including migration, incremental backup, or ABARS backup on data sets in error. Data sets in error could also fail DFSMSdss (DSS) functions such as copy, move, or dump.

Now that we fixed the PDS errors, we can either wait for the next automatic space migration function to migrate the data sets, or we can migrate them using HSM commands interactively from HSM Reporter/Manager panels.

3.3.2 Migrate Data Sets

To migrate the data sets now, key **O** in the 'OPTION 'field and press **Enter**. The *Options* pop-up panel is displayed with all of the options available.

```
Enter Option number
 OPTION==>
   1 SORT
   2 FILTER
   3 PRINT
   4 SAVE
   F FORMAT AND SAVE
   5 PLAN
   6 TOTAL
   7 TAG    (manager)
   8 UNTAG  (manager)
   9 RUN    (manager)
  10 BATCH  (manager)
```

Figure 20: Options Pop-Up Panel

We need to tag the data sets to include in our HSM command. To do this, key **7** (**Tag**) in the 'OPTION' field in the pop-up panel and press **Enter**. The pop-up panel goes away and the letter **T** is automatically placed next to each data set.

Pretend that you only want to migrate the first two data sets and not the third. To de-select the third data set, remove the **T** by tabbing down to the third data set to highlight the **T** and press the space bar. Press **Enter** to return the cursor to the 'Command' line.

```
<V6 R1.xx> HSM REPORTER SELECTED MIGRATE RETURN CODE              Row 1 of 3
OPTION ===>                                                  SCROLL==> PAGE
 ENTER O ON AT OPTION FOR LIST OF OPTIONS.
 PANEL 1 OF 3. SCROLL RIGHT FOR MORE INFORMATION.
 KEY S BESIDE ENTRY FOR RETURN/REASON CODES
S   DSN                                                            ACTION   RC
RSNC  TRKS  AGE
T  GTO.REL315.ADA.NATSOUR.D284              MIGRATE  17 00000    28   36
T  AUD.TUPPER.WAREWL.CAN.TABLE              MIGRATE  17 00000    75   36
T  LMN.FTP.CLWWL4                           MIGRATE  17 00000    75    5
****************************** Bottom of data ********************************
```

Figure 21: 'Un-Tag' A Data Set On The HSM Reporter Selected Migrate Return Code Panel

With the data sets tagged, key **O** in the 'OPTION' field (again) and press **Enter**. The *Options* pop-up panel appears again. This time, key **9** (**Run**) in the 'Command' field, and press **Enter**. The *HSM Manager Line Command Processor* panel is displayed.

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The following is an example of the *HSM Manager Line Command Processor* panel.

```
<V6 R1.xx> HSM REPORTER SELECTED MIGRATE RETURN CODE              Row 1 of 3

                         HSM MANAGER LINE COMMAND PROCESSOR

   HIT PF05 to display LABELS AND DESCRIPTIONS
   HIT PF06 to retrieve a list of commands, HIT PF07 to save this command
   COMMAND=> /SDSN
```

Figure 22: HSM Manager Line Command Processor Panel

This example of the *HSM Manager Line Command Processor* panel (shown above) doesn't show the green command lines you will see on the display, so just imagine that they are there. If you see '/SDSN', or any other command in the 'Command' field, ignore it, we're going to key over it.

The variable we need for the data set names tagged is '/DSN'. We are going to key the HSM command we need to migrate the tagged data sets with /DSN in single quotes to represent the data set names. HSM Reporter/Manager automatically substitutes the tagged data sets names in the command.

For information on the variable names available in each portion of HSM Reporter/Manager, see the *HSM Reporter/Manager Installation & User Guide* (document 002-0202).

Key **`HMIGRATE '/DSN'`** in the 'Command' field. There is a single space between `HMIGRATE` and `'/DSN'`. This is the same command we would use if we issued the migration command from TSO option 6. We enclosed the /DSN in single quotes so HSM doesn't try to append our userid to the front of the data set names.

HSM responds back to our terminal with the following messages.

```
ARC1007I MIGRATE REQUEST 00001257 SENT TO HSM
***
```

Figure 23: HSM response to HMIGRATE '/DSN' request

When HSM responds (with three asterisks), press **Enter** to return to the *HSM Reporter Selected Migrate Return Code* panel. Press **PF3** to exit out of the *HSM Reporter Selected Migrate Return Code* panel.

Refer back to the *HSM Manager Migrate Summary* panel. An example of the *HSM Manager Migrate Summary* panel is shown below.

```
Row 1 of 9
 <V6 R1.xx>       HSM MANAGER MIGRATE SUMMARY
 OPTION ===>                                              SCROLL==> PAGE

                              FOR ALLVOL

 S     RC  COUNT   MESSAGE
       00  000593  MIGRATION OK
     S 17  000003  PDS DIRECTORY READ ERROR
       19  000025  DATA SET IN USE
       30  000004  DATA SET NOT CATALOGED
       37  000044  NO SPACE ON MIG VOL
       51  000001  DATA SET IN USE BY DFHSM
       58  000001  MIGRATION OR DBA DBU FAILED
       70  000015  SMS DS ERROR
       99  002451  UNSUPPORTED DS
 ****************************** Bottom of data ******************************
```

Figure 24: HSM Manager Migrate Summary Panel

The **S** you see in the column next to the return code **17** (not the 'S' - Select column) indicates the last action we keyed in this panel (that's so you can remember where you've been and what you've done).

The next return code (19) means that the data set was in use during the time that space management tried to migrate it. You need to check these data sets every day to see if this is a one-time occurrence or a consistent occurrence. If the data sets are consistently in use, then you need to research why HSM is trying to migrate them. It's common for some data sets to be 'in use' without ever being opened (and thus not updating the last reference date). Another common occurrence is that HSM is attempting to migrate a generation data set but the base is currently being updated and therefore the data set is 'in use (see the IBM 'APAR OW16433' that discusses GDGs and HSM migration activities).

Four data sets received a return code **30**. The description, 'Data Set Not Cataloged', provides enough explanation of the return code.

44 data sets received return code **37** because they could not be migrated due to insufficient space on the migration volume. Tabbed down and selected this error by keying **S** in the 'Select' column and press **Enter**. The *HSM Reporter Selected Migrate Return Code* panel is displayed.

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The following is an example of the *HSM Reporter Selected Migrate Return Code* panel.

```
<V6 R1.xx> HSM REPORTER SELECTED MIGRATE RETURN CODE            Row 1 of 44
 OPTION ===>                                                  SCROLL==> PAGE
  ENTER O ON AT OPTION FOR LIST OF OPTIONS.
  PANEL 1 OF 3. SCROLL RIGHT FOR MORE INFORMATION.
  KEY S BESIDE ENTRY FOR RETURN/REASON CODES
 S   DSN                                          ACTION   RC RSNC  TRKS  AGE
    EXE.LANT.SALING.P20.MASTER1.G0646V00          MIGRATE  37 00000 26400   29
    SYS3.KING.DOME.NOT.HERE.G0001V00              MIGRATE  37 00000 89970   15
    EXE.LANT.SALING.PP.P20.MASTER1.G0484V00       MIGRATE  37 00000 26715   16
    EXE.QUANTUM.LEAP.P20.MASTER1.G0001V00         MIGRATE  37 00000 34755   29
    EXE.TRATER.STREAL.P20.MASTER1.G1187V00        MIGRATE  37 00000 25965   20
    EXE.LANT.SALING.P20.MASTER1.G0653V00          MIGRATE  37 00000 25965   22
    EXE.LANT.SALING.P20.MASTER1.G0675V00          MIGRATE  37 00000 27975   12
    SYS3.KING.DOME.ALL.GONE.G0002V00              MIGRATE  37 00000 88290   15
    EXE.TRATER.STREAL.P20.MASTER1.G1186V00        MIGRATE  37 00000 25965   20
    EXE.LANT.SALING.PP.P40OUT1.UNSORTED.G0246V00 MIGRATE  37 00000 19605   15
    EXE.LANT.SALING.PP.P20.MASTER1.G0492V00       MIGRATE  37 00000 27690   14
    EXE.TRATER.STREAL.P20.MASTER1.G1193V00        MIGRATE  37 00000 25965   17
    EXE.LANT.SALING.P20.MASTER1.G0664V00          MIGRATE  37 00000 26520   17
    EXE.LANT.SALING.P20.MASTER1.G0680V00          MIGRATE  37 00000 38805    8
    EXE.LANT.SALING.PP.P20.MASTER1.G0498V00       MIGRATE  37 00000 28740    9
    EXE.LANT.SALING.P20.MASTER1.G0678V00          MIGRATE  37 00000 28050    9
    EXE.LANT.SALING.P20.MASTER1.G0680V00          MIGRATE  37 00000 38805    8
    EXE.LANT.SALING.PP.P20.MASTER1.G0498V00       MIGRATE  37 00000 28740    9
```

Figure 25: HSM Reporter Selected Migrate Return Code Panel

The example of the *HSM Reporter Selected Migrate Return Code* panel (show above) shows the data set names, the action (migrate), the return code (37), the reason code (0000), the number of tracks, and the age for each data set. You can see why these data sets had trouble migrating to ML1 - *they're BIG*! You can also see how long these data sets have been eligible for migration but could not migrate due to this problem.

This is very easy to fix. Tag them as before and migrate them to ML2 with the following HSM command.

```
HMIGRATE '/DSN' ML2
```

There is a single space between `HMIGRATE` and `'/DSN'` and between `'/DSN'` and `ML2`.

Since there are 44 of these, and you probably have better things to do than press **Enter** every time HSM writes messages to your TSO session, you can do the same commands in batch. The process for building a batch job to process the commands is described in the following topic.

3.3.3 Build A Batch Job to Process HSM Commands

Tag the data sets as before. Key **O** in the 'OPTION' field and press **Enter**. The *Options* pop-up panel is displayed. Key **10** (**Batch**) in the 'OPTION' field and press **Enter**. The *HSM Manager Line Command Processor* pop-up panel will appear. Key the HSM command (HMIGRATE '/DSN' ML2) in the 'Command' field and press **Enter**. The *HSM Manager Control Card Action Menu* panel is displayed.

The following is an example of the *HSM Manager Control Card Action Menu* panel.

```
<V6 R1.xx>        HSM MANAGER CONTROL CARD ACTION MENU
 OPTION  ===> E

    B  BROWSE              -  Browse the file just created
    E  EDIT                -  Edit the file just created
    I  INSTALL             -  Copy file just created to production library
    C  CANCEL              -  Cancel and delete the file just created

 HRO CONTROL CARDS have been created in the data set:
  ===> XX00056.H180610

END (PF3) to Cancel
```

Figure 26: HSM MANAGER CONTROL CARD ACTION MENU panel

In the *HSM Manager Control Card Action Menu* panel, key **E** (Edit) in the 'OPTION' field and press **Enter**.

HSM Reporter/Manager opens a temporary data set that you can modify using standard ISPF Edit commands. To copy the 'TSO IKJEFT01' JCL into this temporary data set, key **COPY** in the 'Command' field, key **A** on the top row of asterisks, and press **Enter**.

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The following is an example of the copy 'TSO IKJEFT01' JCL process.

```
File  Edit  Confirm  Menu  Utilities  Compilers  Test  Help

EDIT       XX00056.H18089                                  Columns 00001 00072
 Command ===>  COPY                                          Scroll ===> CSR
 *A**** ****************************** Top of Data ******************************

 000001  HMIGRATE 'EXE.LANT.SALING.P20.MASTER1.G0646V00' ML2
 000002  HMIGRATE 'SYS3.KING.DOME.NOT.HERE.G0001V00' ML2
 000003  HMIGRATE 'EXE.LANT.SALING.PP.P20.MASTER1.G0484V00' ML2
 000004  HMIGRATE 'CYB.QUANTUM.LEAP.P20.MASTER1.G0001V00' ML2
 000005  HMIGRATE 'EXE.TRATER.STREAL.P20.MASTER1.G1187V00' ML2
 000006  HMIGRATE 'EXE.LANT.SALING.P20.MASTER1.G0653V00' ML2
 000007  HMIGRATE 'EXE.LANT.SALING.P20.MASTER1.G0675V00' ML2
 000008  HMIGRATE 'SYS3.KING.DOME.ALL.GONE.G0002V00' ML2
 000009  HMIGRATE 'EXE.TRATER.STREAL.P20.MASTER1.G1186V00' ML2
 000010  HMIGRATE 'EXE.LANT.SALING.PP.P40OUT1.UNSORTED.G0246V00' ML2
 000011  HMIGRATE 'EXE.LANT.SALING.PP.P20.MASTER1.G0492V00' ML2
 000012  HMIGRATE 'EXE.TRATER.STREAL.P20.MASTER1.G1193V00' ML2
 000013  HMIGRATE 'EXE.LANT.SALING.P20.MASTER1.G0664V00' ML2
 000014  HMIGRATE 'EXE.LANT.SALING.P20.MASTER1.G0680V00' ML2
 000015  HMIGRATE 'EXE.LANT.SALING.PP.P20.MASTER1.G0498V00' ML2
 000016  HMIGRATE 'EXE.LANT.SALING.P20.MASTER1.G0678V00' ML2
```

Figure 27: Example - Copy IKJEFT01 JCL Into Temporary Data Set Opened By HSM Reporter/Manager

After pressing **Enter** in the copy 'TSO IKJEFT01' JCL process, the ISPF Edit/View – COPY panel is displayed. Key in the name of the JCL library and member where you keep things like the 'TSO IKJEFT01' JCL, and press **Enter**.

```
Menu  RefList  Utilities  Help
Edit/View - Copy
 Command ===>
                                                                    More:     +
 "Current" Data Set: XX00056.H18089

 From ISPF Library:
    Project . . .
    Group . . . .          . . .          . . .          . . .
    Type  . . . .
    Member  . . .               (Blank or pattern for member selection list)

 From Other Partitioned or Sequential Data Set:
    Data Set Name  . . 'XX00056.JCL.CNTL(TSO)'
    Volume Serial  . .          (If not cataloged)

 Data Set Password  . .         (If password protected)

 Line Numbers (Blank for entire member or sequential data set)
    First line  . . . .
    Last line . . . . .
    Number type . . . .         (Standard, ISPFstd, COBOL, or Relative)
```

Figure 28: ISPF Edit/View – COPY Panel

The JCL is automatically copied into the temporary data set. The following is an example of this process.

```
File  Edit  Confirm  Menu  Utilities  Compilers  Test  Help
EDIT       XX00056.H18089                                MEMBER TSO copied
 Command ===>                                              Scroll ===> CSR
 ****** ***************************** Top of Data ******************************
 000001 //XX00056G JOB (0-7409),'COLLEEN GORDON',
 000002 //         CLASS=X,MSGCLASS=1,REGION=4096K,NOTIFY=SUBMITR
 000003 //TSCMD  EXEC PGM=IKJEFT01
 000004 //SYSTSPRT DD SYSOUT=*
 000005 //SYSPRINT DD SYSOUT=*
 000006 //SYSIN DD DUMMY
 000007 //SYSTSIN DD *
 ==MSG> -Warning- The UNDO command is not available until you change
 ==MSG>           your edit profile using the command RECOVERY ON.
000009  HMIGRATE 'EXE.LANT.SALING.P20.MASTER1.G0646V00' ML2
 000010  HMIGRATE 'SYS3.KING.DOME.NOT.HERE.G0001V00' ML2
 000011  HMIGRATE 'EXE.LANT.SALING.PP.P20.MASTER1.G0484V00' ML2
 000012  HMIGRATE 'CYB.QUANTUM.LEAP.P20.MASTER1.G0001V00' ML2
 000013  HMIGRATE 'EXE.TRATER.STREAL.P20.MASTER1.G1187V00' ML2
 000014  HMIGRATE 'EXE.LANT.SALING.P20.MASTER1.G0653V00' ML2
 000015  HMIGRATE 'EXE.LANT.SALING.P20.MASTER1.G0675V00' ML2
 000016  HMIGRATE 'SYS3.KING.DOME.ALL.GONE.G0002V00' ML2
 000017  HMIGRATE 'EXE.TRATER.STREAL.P20.MASTER1.G1186V00' ML2
```

Figure 29: TSO IKJEFT01 JCL Copied Into ISPF Edit Panel

If you get that warning message in your JCL (like we did in our example), key **RES** (Restore) in the 'Command' field and press **Enter** to make the warning go away.

Now you're ready to submit this as a batch job. Double check your job card, key **SUB** (Submit) in the 'Command' field and press **Enter**. We usually split the screen here (**PF2**) and check the job's progress in SDSF (or equivalent) to make sure everything ran OK. All you will see in the SYSOUT is that the command was sent to HSM. If you get a message that the command failed, you did something wrong. Go back and try it again.

You're done with the temporary data set. Press **PF3** to exit out of this panel. The *HSM Manager Control Card Action Menu* is displayed. Press **PF3** to exit out of this menu. The *HSM Reporter Selected Migrate Return Code* panel is displayed. Press **PF3** again to exit out of the *HSM Reporter Selected Migrate Return Code* panel.

There's one more thing that I want to talk about. It's the 2,451 unsupported data sets that could not migrate that we saw on the *Migrate Summary* panel (page 22).

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3.3.4 Unsupported Data Sets

Let's take a look at the unsupported data sets by selecting on return code **99**. The *HSM Reporter Selected Migrate Return Code* panel is displayed.

The following is an example of the *HSM Reporter Selected Migrate Return Code* panel.

```
<V6 R1.xx> HSM REPORTER SELECTED MIGRATE RETURN CODE            Row 1 of 4323
 OPTION ===>                                                  SCROLL==> PAGE
  ENTER O ON AT OPTION FOR LIST OF OPTIONS.
  PANEL 1 OF 3. SCROLL RIGHT FOR MORE INFORMATION.
  KEY S BESIDE ENTRY FOR RETURN/REASON CODES
 S   DSN                                          ACTION   RC RSNC  TRKS  AGE
 S  TEST.THE.OCT00T.HORPE.G0050V00                MIGRATE  99 00004     0  113
    LMN.TART.LEV.MON1.BOMHDR2.NUT.G0316V00        MIGRATE  99 00004     0   21
    LMN.TART.LEV.CELL2.BOMHDR2.NUT.G0410V00       MIGRATE  99 00004     0    6
    ACE.TSTG00.ALPACA7                            MIGRATE  99 00004     0    5
    EAT.DOVEBAR.V6X0.PGM042.REPROC.TST600         MIGRATE  99 00004     0    2
    LMN.TART.LEV.KRSPEC.BOMHDR2.NUT.G0032V00      MIGRATE  99 00004     0   58
    ACE.TSTG00.ALPACA2                            MIGRATE  99 00004     0    5
    APT.TS3.P81.APT426.OUTP.G0005V00              MIGRATE  99 00004     0   82
    APT.TS2.CG.INV                                MIGRATE  99 00004     0   63
    GTO.HLS.TEST.C0412CH.WORK13                   MIGRATE  99 00004     0   48
    EXE.ST30TEST.SLYPTY.QUERY10.G0200V00          MIGRATE  99 00004     0   82
    EXE.ST30TEST.SLYPTY.QUERY10.G0204V00          MIGRATE  99 00004     0   78
    NC99706.DOUNOW.STEP1.SRTOUT                   MIGRATE  99 00004     0    7
    NC99208.PLEATHER.UNLOADNW.SYSREC38            MIGRATE  99 00004     0   82
    ACE.G00.ALPACA7                               MIGRATE  99 00004     0    1
    TEST.THE.OCT00T.HORPE.G0050V00                MIGRATE  99 00004     0  113
    LMN.TART.LEV.MON1.BOMHDR2.NUT.G0316V00        MIGRATE  99 00004     0   21
    LMN.TART.LEV.CELL2.BOMHDR2.NUT.G0410V00       MIGRATE  99 00004     0    6
```

Figure 30: HSM Reporter Selected Migrate Return Code Panel Displaying Unsupported Data Sets

Use the 'Tab' key to tab down to the first data set name and key **S** in the 'Select' column next to the data set name. The *HSM Reporter Return Code Reason Code Display* pop-up panel is displayed.

The following is an example of the *HSM Reporter Return Code Reason Code Display* pop-up panel.

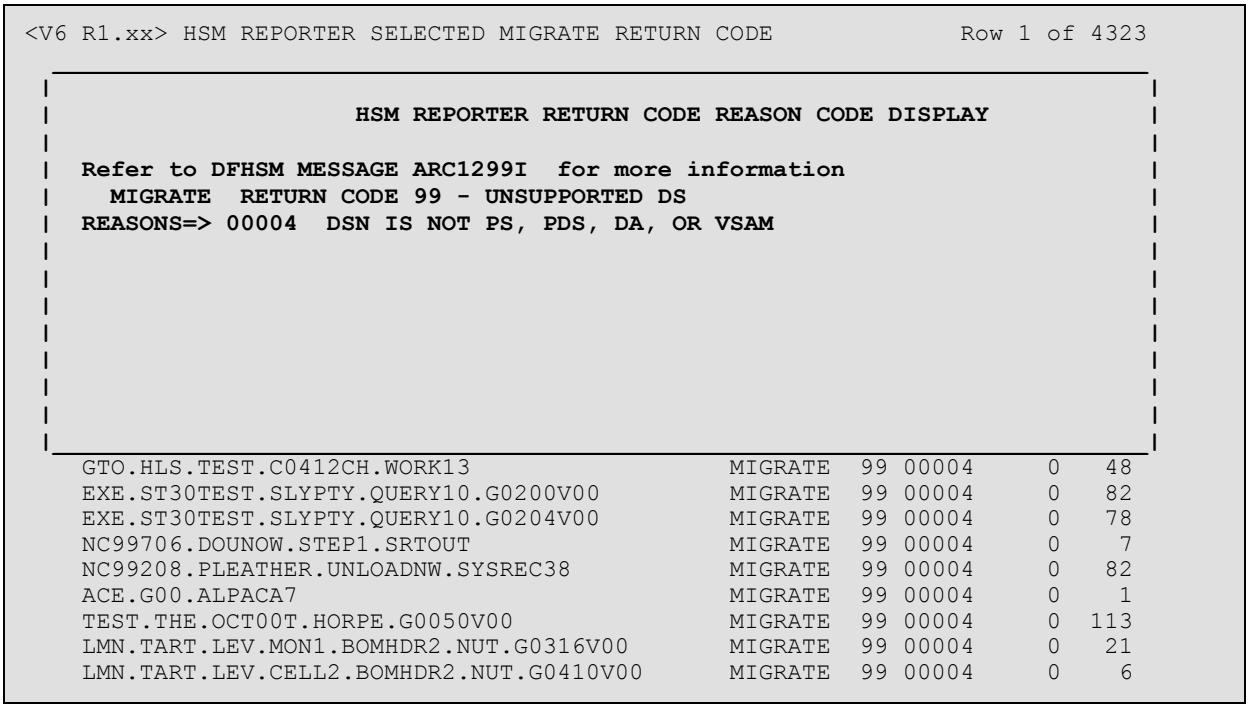


Figure 31: HSM Reporter Return Code Reason Code Display Pop-Up Panel

This panel is very useful because it provides the HSM ARC message 'ARC1299I' (for this example) and the return code (if any) so you can look it up in the messages and codes manual. It also provides a text message 'DSN IS NOT PS, PDS, DA OR VSAM' (for this example) letting you know what the error is. Most of the time, you can figure out what's wrong just from reading this message and you won't need to look up the ARC message in the manual.

These errors are the result of an application creating (allocating) a data set but never opening it. The DSORG is not assigned until the data set is opened. This also means that all of these data sets are empty; because they were never opened for write. We have also shown how quickly the number of data sets with DSORG unknown grows, leaving thousands of data sets on your primary DASD for the life of the data sets.

3.3.5 How To Never See DSORG Unknown Again!

These errors are easily preventable by assigning an SMS data class to all data sets in your shop. You can assign a data class with NO attributes (if you desire) to both SMS managed and non-managed data. Because you assign a data class, SMS assigns a DSORG regardless of whether the data set was opened for write. If you would like more information on assigning a data class to all data sets in your shop, or you would like to see some examples of how this is coded in SMS ACS routines, see the data class articles, Part I and Part II provided in *Appendix A, Data Class Articles*.

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3.3.6 Display Migrate/Backup Totals

Now, back to business. The 4,000+ data sets currently cataloged can not be migrated, or backed up (by incremental backup). All you can do is let them die their natural death. You can certainly use this information to show management how much space these data sets occupy on primary DASD, giving you plenty of financial reasons to implement data class in your shop.

Here's how:

In the *HSM Reporter Selected Migrate Return Code* panel (this is the one we were just in), key **TOTAL** in the 'Command' field and press **Enter**. The *HSM Manager Migrate/Backup Display Totals* panel is displayed.

The following is an example of the *HSM Manager Migrate/Backup Display Totals* panel.

```
<V6 R1.xx>          HSM MANAGER MIGRATE/BACKUP DISPLAY TOTALS
 COMMAND ===>


  NUMBER OF ENTRIES IN TABLE==> 54


  LABEL                TOTAL     AVERAGE     MINIMUM     MAXIMUM
                     ---------     ------     -------     -------
  TRACKS ======>     1868685       34605       25980       89970
  AGE =========>         619          11           5          21
  CPU(.01 SEC)=>           0           0           0           0



 ENTER END TO TERMINATE
```

Figure 32: HSM Manager Migrate/Backup Display Totals Panel

Press **PF3** to exit out of this panel.

Fro the above panel, you can easily report on the amount of storage that is wasted by this problem. **TOTAL** is a great command that can be used from all HSM Reporter/Manager panels that display data set information. This summary is also provided in printed reports. The process to generate a printed report is described in the following topic.

3.3.7 Print Reports

In the same panel (*HSM Reporter Selected Migrate Return Code*), where we keyed **TOTAL** in the 'Command' field. Key **O** (**OPTIONS**) in the 'Command' field and press **Enter**. You will see the *Options* pop-up panel. Key **3** (**Print**) in the 'OPTION' field and press **Enter**.

```
<V6 R1.xx> HSM REPORTER SELECTED MIGRATE RETURN CODE      Row 1 to 18 of 712
 OPTION ===> O       |--------------------------|            SCROLL==> PAGE
  ENTER O ON AT OPTI | Enter Option number      |
  PANEL 1 OF 3. SCRO | OPTION==>                |
  KEY S BESIDE ENTRY |   1 SORT                 |
 S   DSN             |   2 FILTER               | CTION   RC RSNC  TRKS  AGE
    $IAM.LOGREC2.FOR |   3 PRINT                | IGRATE  00 00000    15    3
    $IAM.J7867140.FO |   4 SAVE                 | IGRATE  00 00000     2    3
    $IAM.J773IU1T.FO |   5 FORMAT AND SAVE      | IGRATE  00 00000     1    3
    $IAM.J773IU2T.FO |   5 PLAN                 | IGRATE  00 00000     1    3
    $IAM.J764S01D.FO |   6 TOTAL                | IGRATE  00 00000     2    3
    $IAM.J773IU1T.FO |   7 TAG   (manager)      | IGRATE  00 00000     1    3
    $IAM.J773DW01.FO |   8 UNTAG (manager)      | IGRATE  00 00000     2    3
    $IAM.J3877824.FO |   9 RUN   (manager)      | IGRATE  00 00000     2    3
    $IAM.J773IU2T.FO |  10 BATCH (manager)      | IGRATE  00 00000     2    3
    $IAM.J221SE01.FO |--------------------------| IGRATE  00 00000     2    3
    $IAM.J3871603.FORMC060.D000415.T012413       MIGRATE  00 00000    11    3
    $IAM.J3875910.FORMR323.D000415.T143511       MIGRATE  00 00000     1    3
    $IAM.J773IU2T.FORMRWF2.D000415.T121146       MIGRATE  00 00000     2    3
    $IAM.J970MINR.FORMC099.D000415.T012501       MIGRATE  00 00000     3    3
    $IAM.J788BD03.FORMV781.D000415.T003231       MIGRATE  00 00000     1    3
    $IAM.J788BD03.FORMV432.D000415.T003226       MIGRATE  00 00000     2    3
    $IAM.J236ER01.FORMS085.D000415.T003648       MIGRATE  00 00000     2    3
    $IAM.X3879581.FORMI053.D000415.T154802       MIGRATE  00 00000    18    3
```

Figure 33: HSM Reporter Selected Migrate Return Code Panel With The Options Pop-Up Panel

HSM Reporter/Manager generates a batch job that creates the report. Use the JOBCARD you coded into the *JCL Parameters* panel, so the output goes to the MSGCLASS parameter you coded. Press **PF3** to exit out of this panel.

3.3.8 Logfile Migration/Backup Option 2 - ML1 to ML2 Return Code Summary

Now this time we will look at ML1 to ML2 return codes. Go back to the *HSM Manager Logfile Migration/Backup Options* panel by pressing **PF3** one more time.

Key **2** in the 'Command' field and press **Enter**. The *HSM Manager MIG1TO2 Summary* panel is displayed.

The following is an example of the *HSM Manager MIG1TO2 Summary* panel.

```
Row 1 to 1 of 1
<V6 R1.xx>       HSM MANAGER MIG1TO2 SUMMARY
OPTION ===>                                              SCROLL==> PAGE

                            FOR ALLVOL

S    RC  COUNT   MESSAGE
     00  000319  MIGRATION OK
******************************* Bottom of data ********************************
```

Figure 34: HSM Manager MIG1TO2 Summary Panel Shows ML1 To ML2 Return Codes

Today there weren't any problems with this aspect of space management.

To look at the detail data sets that were migrated, tab down to the 'Select' column, key **S** (for an RC), and press **Enter**. The *HSM Reporter Selected MIG1TO2 Return Code* panel is displayed.

The following is an example of the *HSM Reporter Selected MIG1TO2 Return Code* panel.

```
<V6 R1.xx> HSM REPORTER SELECTED MIG1TO2 RETURN CODE    Row 1 to 18 of 2,573
 OPTION ===>                                              SCROLL==> PAGE
  ENTER O ON AT OPTION FOR LIST OF OPTIONS.
  PANEL 1 OF 3. SCROLL RIGHT FOR MORE INFORMATION.
  KEY S BESIDE ENTRY FOR RETURN/REASON CODES
 S   DSN                                          ACTION   RC RSNC  TRKS  AGE
    ACE.ANT243.CHKREG.EXTRACT.G0271V00            MIG1TO2  00 00000     1   20
    ACE.ANT304.FILE.G1449V00                      MIG1TO2  00 00000     1   20
    ACE.ANT304.FILE.G1450V00                      MIG1TO2  00 00000     1   20
    ACE.ANT390.FILE.G0886V00                      MIG1TO2  00 00000     1   20
    ACE.CHKREG.G1636V00                           MIG1TO2  00 00000    26   20
    ACE.CLERK.TOTALS.BKUP.G0031V00                MIG1TO2  00 00000     1   20
    ACE.DIV.TOTALS.BKUP.G0031V00                  MIG1TO2  00 00000     1   20
    ACE.ANT304.FILE.SORTED.G0028V00               MIG1TO2  00 00000     1   20
    ACE.FAXINP.G0173V00                           MIG1TO2  00 00000     1   20
    ACE.FRT.PAYMENT.CAN.US.G0106V00               MIG1TO2  00 00000     1   20
    ACE.FRT.PAYMENT.G0273V00                      MIG1TO2  00 00000     1   20
    ACE.FRT.PAYMENT.G0274V00                      MIG1TO2  00 00000     1   20
    ACE.FREDDI.DETAIL.BKUP.G0256V00               MIG1TO2  00 00000     1   20
    ACE.INV.VENTURA.DOG.TRAINER.G2018V00          MIG1TO2  00 00000     5   20
    ACE.INV.VENTURA.CAN.US.TRAINER.G0867V00       MIG1TO2  00 00000     1   20
    ACE.INV.VENTURA.CAN.US.TRAINER.TWO.G1604V00   MIG1TO2  00 00000     1   20
    ACE.INV.VENTURA.CAN.US.TRAINER.TWO.G1605V00   MIG1TO2  00 00000     1   20
    ACE.VENTURA.PET.DETECTV.G0224V00              MIG1TO2  00 00000     1   20
    ACE.VENTURA.PET.DETECTV.G0224V00              MIG1TO2  00 00000     1   20
```

Figure 35: HSM Reporter Selected MIG1TO2 Return Code Panel

You can scroll right on this panel to show more information. This information gives you an idea of how much data is migrated to ML2 every day.

We will look at the HSM Incremental Backup return codes next. Press **PF3** to exit out of this panel. Press **PF3** one more time to return to the *HSM Manager Logfile Migration/Backup Options* panel. .

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3.3.9 Logfile Migration/Backup Option 3 - Backup Return Code Summary

If you are using HSM Incremental Backup, you can check out the success or failure of backups within the reporting period (normally the last 24 hours). To look at the backup return code summary, key **3** in the 'Command' field of the *HSM Manager Logfile Migration/Backup Options* panel and press **Enter**. The *HSM Manager Backup Summary* panel is displayed.

The following is an example of the *HSM Manager Backup Summary* panel.

```
Row 1 to 5 of 5
<V6 R1.xx>       HSM MANAGER BACKUP SUMMARY
OPTION ===>                                              SCROLL==> PAGE

                             FOR ALLVOL

S    RC  COUNT   MESSAGE
     00  004881  BACKUP OK
     03  000001  FMT 1 DSCB OBTAIN ERROR
     04  000010  CANT MOVE BKUP, MIGVOL NOT AVAIL
     19  000251  DATA SET IN USE
     99  000001  UNSUPPORTED DS
****************************** Bottom of data ********************************
```

Figure 36: HSM Manager Backup Summary Panel

You may notice similar types of failures in the *Backup Summary* panel as we saw in the *Migration Summary* panel. This is due to the fact that if HSM can not migrate the data set due to an error, it can not back it up either. When HSM incremental backup and migration is used for data sets in an SMS management class, the backup must be performed prior to the migration function. It is common for a data set to fail migration because the backup failed.

Like with migration errors, backup errors need to be resolved and the backup function retried. In the *HSM Manager Backup Summary* panel (shown above), we can see data sets that failed because of PDS read errors, data sets that are not cataloged, data sets that failed due to DSS errors, and more. We won't discuss how to fix these types of errors, but once you get them resolved, you can certainly use the panel to issue the HSM backup commands either in foreground or in batch, just like we did with the migration errors. Press **PF3** to return to the *HSM Manager Master Panel*.

3.4 Recall/Recover Options - Master Panel Option 3

Another great feature of HSM Reporter/Manager is the ability to look at the data sets that are getting recalled back to DASD. You can monitor these recalls to see if you need to change your SMS ACS routines to reduce the number of recalls. Recalling a data set, either from DASD or from tape, has a considerable cost due to the CPU cycles required to decompress the data. It is now recommended not to compress data sets going to ML2, especially if you already have a compression tool such as IDRC. Data sets can fail recall due to errors, so you need to check the recall summary daily and fix those problems too.

To look at the recalled data sets, key **3** in the 'Command' field on the *HSM Reporter Master Panel* and press **Enter**. The *HSM Manager Logfile Recall/Recover Options panel* is displayed. Key **1** in the 'Command' field and press **Enter**. The *HSM Manager Recall Summary* panel is displayed.

The following is an example of the *HSM Manager Recall Summary* panel.

```
Row 1 to 4 of 4
 <V6 R1.xx>       HSM MANAGER RECALL SUMMARY
 OPTION ===>                                              SCROLL==> PAGE

                            FOR ALLVOL

 S     RC  COUNT   MESSAGE
       00  000981  RECALL:RECOVERY OK
       02  000557  DS NOT MIGRATED
       51  000468  OTHER FUNCTION USING DATA SET
       82  000005  TAPE VOL NOT AVAIL
 ******************************** Bottom of data ********************************
```

Figure 37: HSM Manager Recall Summary Panel

In the *HSM Manager Recall Summary panel* (shown above), we can see that a couple of data sets have failed recall due to different reasons. These panels work the same way as the migration and backup panels, so everything you have already learned thus far can be applied here.

Once you fix all the problems, you can reissue the RECALL command using the panels either in foreground or in batch (if there's a bunch). We usually like to test that the fix we applied worked by attempting the recall again. Press **PF3** twice (to exit out of this panel and the next one) to return to the *HSM Manager Master Panel*.

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3.5 Extent Reduction Summary - Master Panel Option 4

If you are using HSM to reduce extents, you can take a look at those records using HSM Reporter/Manager as well. Key **4** in the 'Command' field of the *HSM Manager Master Panel* and press **Enter**. The *HSM Reporter Online Extent Reduction Selection* panel is displayed.

The following is an example of *the HSM Reporter Online Extent Reduction Selection* panel.

```
<V6 R1.xx>     HSM REPORTER ONLINE EXTENT REDUCTION SELECTION

 COMMAND==>

 SELECT A CATAGORY OF EXTENT REDUCTION BELOW.


        1  MIGRATION FOR EXTENT REDUCTION RETURN CODES
        2  RECALL FOR EXTENT REDUCTION RETURN CODES
        3  EXTENT REDUCTION RESULTS

  Enter  END  To terminate online function
```

Figure 38: HSM Reporter Online Extent Reduction Selection Panel

There are three options available on the *HSM Reporter Online Extent Reduction Selection* panel.

1. Migration for Extent Reduction Return Codes.
2. Recall for Extent Reduction Return Codes.
3. Extent Reduction Results.

Why all these options?

To perform extent reduction, HSM needs to migrate the data set to ML1 then recall it back to DASD. It uses HSM's compression algorithms to reduce the extents. Since this process is CPU intensive, it's recommended that you adjust the 'MAXEXTENTS(NN)' parameter in HSM's PARMLIB to a number that provides the best value for the cost. This number is going to be different for everybody. Some shops are going with 20 extents.

3.5.1 Extent Reduction Option 1 - Migration For Extent Reduction Return Codes

To look at the migration for extent reduction return codes, key **1** in the 'Command' field of the *HSM Reporter Online Extent Reduction Selection* panel and press **Enter**. The *HSM Manager Reduced Summary* panel is displayed.

The following is an example of the *HSM Manager Reduced Summary* panel.

```
                                                                  Row 1 to 3 of 3
<V6 R1.xx>           HSM MANAGER REDUCED SUMMARY
 OPTION ===>                                                   SCROLL==> PAGE

                                    FOR ALLVOL

 S    RC  COUNT   MESSAGE
      00  000020  MIGRATION OK
      19  000002  DATA SET IN USE
      37  000005  NO SPACE ON MIG VOL
************************************ Bottom of data ************************
```

Figure 39: HSM Manager Reduced Summary Panel

In the *HSM Manager Reduced Summary* panel (shown above), we have some data sets that failed extent reduction. These need to be fixed before extent reduction can be retried. Press **PF3** to exit out of this panel.

3.5.2 Extent Reduction Option 2 - Recall For Extent Reduction Return Codes

Usually, if the migration for extent reduction was successful, the recall is successful as well. But to be sure, key **2** in the 'Command field' on the *HSM Reporter Online Extent Reduction Selection* panel and press **Enter**. When you are done, press **PF3** to exit out of this panel.

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3.5.3 Extent Reduction Option 3 - Extent Reduction Results

Now we will look at the extent reduction results. Key **3** in the 'Command' field on the *HSM Reporter Online Extent Reduction Selection* panel and press **Enter**. The *HSM Manager Extent Reduction Summary* panel is displayed.

The following is an example of the *HSM Manager Extent Reduction Summary* panel.

```
                                                                  Row 1 to 12 of 20
<V6 R1.xx>           HSM MANAGER EXTENT REDUCTION SUMMARY
 OPTION ===>                                                   SCROLL==> PAGE
  VALID OPTIONS ARE SORT TO SORT THE TABLE, P TO PRINT THE TABLE.
SPECIFY SORTKEY===>             OVOL, KB2, TKW, TKR RC OR DSN
SPECIFY SORTTYPE==>             A FOR ASCENDING, D FOR DESCENDING
 PRIMARY KB2    TRKS  TRKS  RECALL
 VOL     BYTES  MIGR  RCLL  RC     DSNAME
 BTH035  082904 03045 03649 OO M342.T3U.EB.STAMPED.I766TRAN.G1433V00
 BTH038  008562 00318 00408 OO M763.P2000000.DEXTRACT.G1682V00
 BTH040  003893 00144 00186 OO M939.M753GT.SEQ03115
 BTH073  090332 03337 04303 OO M239.P2030.MKTGTRNS.BACKUP.G0181V00
 BTH042  001382 00051 00066 OO M112.P2502007.SORTED.CODED.G1468VOO
 BTH053  001756 00353 00084 OO M965.P1015.SP.DREPORT1.G0320V00
 BTH048  001845 00081 00088 00 M909.PJOURISD.TAJRPT1
 BTH075  088452 03885 04214 OO DUMPD.EBDM401D.DALY.INVUP03.G1413V00
 BTH075  009912 03371 00473 OO M648.P2025001.PROM0175
 BTH075  008820 00325 00421 OO M016.P2025006.PROM0108
 BTH075  008820 00325 00421 OO M931.P2025008.PROM0108
 BTH075  008820 00325 00421 OO M126 P2025010.PROM0108.G0414V00
```

Figure 40: HSM Manager Extent Reduction Summary Panel

This panel is really helpful because it gives you all the information about the data set's size and what volume it lives on. You can see how many tracks the data set occupies in migration, as well as how many tracks it occupies on primary DASD. The *HSM Manager Extent Reduction Summary* panel (shown above) is an example of how much space is saved when the data set is migrated to ML1. Press **PF3** twice (to exit out of this panel and the next one) to return to the *HSM Manager Master Panel*.

Now we can look at some more good stuff.

3.6 Recycle Return Code Summary - Master Panel Option 5

Option 5 on the *HSM Manager Master Panel* is the recycle return code summary. You can check on the progress of HSM's recycle using this option. Key **5** in the 'Command' field and press **Enter**. The *HSM Manager Recycle Summary* panel is displayed.

The following is an example of the *HSM Manager Recycle Summary* panel.

```
Row 1 to 1 of 1
 <V6 R1.xx>       HSM MANAGER RECYCLE SUMMARY
 OPTION ===>                                              SCROLL==> PAGE

                             FOR ALLVOL

 S    RC  COUNT   MESSAGE
      00  000086  ALL DATA SETS RECYCLED OK
 ******************************* Bottom of data ********************************
```

Figure 41: HSM Manager Recycle Summary Panel

Tab down to the 'Select column' on the *HSM Manager Recycle Summary*, key **S,** and then press **Enter**. The *HSM Manager Recycle Activity* panel is displayed.

The following is an example of the *HSM Manager Recycle Activity* panel.

```
Row 1 to 17 of 56
 <V6 R1.xx>       HSM MANAGER RECYCLE ACTIVITY
 OPTION ===>                                              SCROLL==> PAGE
  ENTER A NON BLANK CHARACTER IN THE S FIELD TO ZOOM IN FOR MORE DETAIL
                RC=00 MESSAGE=ALL DATA SETS RECYCLED OK
          DSNS   RESN  SYSTEM   END      END      START    START    ELAPSED
 S FVOL   RCYCLD CODE  ID       DATE     TIME     DATE     TIME     TIME
   606141     31 00000 UNKNOWN  2000109 04:02:03 ....... ..:..:.. ..:..:..
   602539     15 00000 NWL1     2000109 04:15:34 2000109 04:02:03 00:13:31
   601291    327 00000 UNKNOWN  2000109 04:03:33 ....... ..:..:.. ..:..:..
   601322     80 00000 NWL1     2000109 04:10:13 2000109 04:03:33 00:06:40
   603566    506 00000 UNKNOWN  2000109 04:05:48 ....... ..:..:.. ..:..:..
   601214     16 00000 NWL1     2000109 04:18:20 2000109 04:05:49 00:12:31
   604381    623 00000 UNKNOWN  2000109 04:06:38 ....... ..:..:.. ..:..:..
   610357    335 00000 NWL1     2000109 04:15:45 2000109 04:06:39 00:09:06
   601295     11 00000 UNKNOWN  2000109 04:09:10 ....... ..:..:.. ..:..:..
   601307      2 00000 NWL1     2000109 04:13:18 2000109 04:09:10 00:04:08
   604960    128 00000 NWL1     2000109 04:18:26 2000109 04:10:13 00:08:13
   600941    277 00000 NWL1     2000109 04:26:17 2000109 04:15:45 00:10:32
   602419     13 00000 NWL1     2000109 04:19:03 2000109 04:18:20 00:00:43
   606470     23 00000 NWL1     2000109 04:31:56 2000109 04:18:27 00:13:29
```

Figure 42: HSM Manager Recycle Activity Panel

The *HSM Manager Recycle Activity* panel also contains helpful information. You can see which volumes were recycled, the number of data sets recycled, the start time, end time, and the elapsed time. To look at more information, tab down next to the 'FVOL' VOLSER, key **S** in the 'Select column', and press **Enter**. You can now see all of the information available for this volume including the data set names, return code with text, reason code with text, from and to volume, and the date and time of the event. Press **PF3** twice to exit out of this panel.

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3.7 Return Code Reason Code Tables - Master Panel Option 6

Option 6 from the *HSM Manager Master Panel* is a great source of information regarding the return and reason codes HSM uses for migration, backup, recall, and recover. Key **6** in the 'Command' field and press **Enter**. The *HSM Manager Return Code Reason Code Panel* is displayed.

The following is an example of the *HSM Manager Return Code Reason Code Panel*.

```
<V6 R1.xx>     HSM MANAGER RETURN CODE REASON CODE PANEL

COMMAND==>

THIS IS THE HSM REPORTER ONLINE RETURN CODE TABLES.  SELECT A CATEGORY

       1  MIGRATION RETURN CODES
       2  BACK-UP RETURN CODES
       3  RECALL RETURN CODES
       4  RECOVER RETURN CODES

 enter  END  To terminate online function
```

Figure 43: HSM Manager Return Code Reason Code Panel

To look at the table for migration return codes, key **1** in the 'Command field' and press **Enter**. The *HSM Manager Migrate Return Code Panel* is displayed.

The following is an example of the *HSM Manager Migrate Return Code Panel*.

```
Row 1 to 18 of 100
<V6 R1.xx>      HSM MANAGER MIGRATE RETURN CODE PANEL
OPTION ===>                                          SCROLL==> PAGE
 ENTER S IN THE S FIELD OR ENTER RC NUMBER ON COMMAND LINE FOR
 REASON CODES
S     RC  MESSAGE
      00  MIGRATION OK
      01  UNKNOWN ERROR TYPE
      02  CATALOG LOCATE ERROR
      03  FMT 1 DSCB OBTAIN ERROR
      04  DATA SET NOT MOVABLE
      05  NO MIGRATION VOLUME AVAILABLE
      06  DUPLICATE DSN IN MCDS
      07  FMT 3 DSCB OBTAIN ERROR
      08  MIG COPY ALLOCATION ERROR
      09  MIG COPY RENAME ERROR
      10  VSAM MCA RECORD EXISTS
      11  MCDS RECORD UPDATE ERROR
      12  MIGRATION COPY ALREADY EXISTS
      13  SYSTEM CAT UPDATE ERROR
      14  DS SCRATCH ERROR
      15  PDS MULTIPLE NOTELIST ERROR
      16  PRIMARY COPY READ ERROR
      17  PDS DIRECTORY READ ERROR
```

Figure 44: HSM Manager Migrate Return Code Panel

To see all of the reason codes for each return code, tab down to the S (Select) column, key **S,** and press **Enter**. You can check the BACKUP (**option 2**), RECALL (**option 3**), and RECOVER (**option 4**) Return and Reason Codes using the same procedure. Press **PF3** until you return to the *HSM Reporter Master Panel*.

3.8 Log Message Summary - Master Panel Option 7

From time to time you will need to check out messages that are written to the console log for a variety of reasons. For example, you need to know what time HSM space migration started and ended. **Option 7** from the *HSM Manager Master Panel* gives you all those great messages. Key **7** in the 'Command field' and press **Enter**. The *HSM Manager Logfile Browse Selection* panel is displayed.

The following is an example of the *HSM Manager Logfile Browse Selection* panel.

```
<V6 R1.xx>     HSM MANAGER LOGFILE BROWSE SELECTION

 COMMAND==>

 SPECIFY LOGFILE BROWSE SELECTION BELOW:

        1  SUMMARIZE LOG MESSAGES
        2  BROWSE ALL MESSAGES
        3  SUMMARIZE BY DFHSM FUNCTIONS
        4  FILTER MESSAGES
        5  BUILD A MESSAGE FILTER PLAN
        6  DISPLAY MESSAGE FILTER PLANS
        7  DISPLAY FILTERED MESSAGE FLAT FILES

  enter  END  To terminate online function
```

Figure 45: HSM Manager Logfile Browse Selection Panel

As you can see, the *HSM Manager Logfile Browse Selection* panel provides several options to choose from. Using the above example of looking for the start and end time of HSM's space migration, you need to look at **option 3** (SUMMARIZE BY DFHSM FUNCTION).

Key **3** in the 'Command' field of the *HSM Manager Logfile Browse Selection* panel and press **Enter**. The *HSM Manager Function Summary Browse* panel is displayed.

The following is an example of the *HSM Manager Function Summary Browse* panel.

```
<V6 R1.xx>     HSM MANAGER FUNCTION SUMMARY BROWSE

 COMMAND==>
 SPECIFY FUNCTION TYPE BELOW:
  A P IN FRONT OF THE SELECTION NUMBER WILL CAUSE A BATCH PRINT
  OF THE SELECTED HSM LOGGED MESSAGES, E.G. P0, P1, etc.

        0  FUNCTION SUMMARY REPORT
        1  PRIMARY SPACE MANAGER DETAIL   (DFHSM 2.6 OR ABOVE)
        2  SECONDARY SPACE MANAGER DETAIL (DFHSM 2.6 OR ABOVE)
        3  DAILY SPACE MANAGER            (DFHSM 2.5 OR BELOW)
        4  MIGRATION CLEAN-UP
        5  AUTO BACK-UP MIGRATED DATA SETS
        6  AUTO BACK-UP VOLUMES
        7  AUTO BACK-UP CONTROL DATA SETS
        8  AUTO DUMP VOLUMES
        9  RECYCLE MESSAGES

  enter  END  To terminate online function
```

Figure 46: HSM Manager Function Summary Browse Panel

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Key **1** in the 'Command field' of the *HSM Manager Function Summary Browse* panel and press **Enter**.

A temporary data set is created by HSM Reporter/Manager and all of the log messages are provided for you to view:

```
BROWSE    SSGSXT.MSGZOOM                              Line 00000000 Col 001 080
Command ===>                                            Scroll ===> PAGE
******************************** Top of Data ********************************
           HSM REPORT ONLINE LIST OF SELECTED  LOG MESSAGES <V5R2.01>
                      REPORT DATE AND TIME 2000109 17:05:00
000108 190000**OPER**.ARC0520I PRIMARY SPACE MANAGEMENT STARTING $$$$NC50
000108 190000**OPER**$ARC0400I VOLUME HSMM01 IS 39% FREE, 001660 FREE TRACK(S),
000108 190000**OPER**âARC0401I LARGEST EXTENTS FOR HSMM01 ARE CYLINDERS    217
000108 190001**OPER**$ARC0400I VOLUME HSMM02 IS 36% FREE, 001492 FREE TRACK(S),
000108 190001**OPER**âARC0401I LARGEST EXTENTS FOR HSMM02 ARE CYLINDERS    146
000108 190001**OPER**$ARC0400I VOLUME HSMM04 IS 37% FREE, 001903 FREE TRACK(S),
000108 190001**OPER**âARC0401I LARGEST EXTENTS FOR HSMM04 ARE CYLINDERS    120
000108 190001**OPER**$ARC0400I VOLUME HSMM05 IS 31% FREE, 001363 FREE TRACK(S),
000108 190001**OPER**âARC0401I LARGEST EXTENTS FOR HSMM05 ARE CYLINDERS    174
000108 190002**OPER**$ARC0400I VOLUME HSMM06 IS 35% FREE, 001357 FREE TRACK(S),
000108 190002**OPER**âARC0401I LARGEST EXTENTS FOR HSMM06 ARE CYLINDERS    461
000108 190002**OPER**$ARC0400I VOLUME HSMM07 IS 37% FREE, 001159 FREE TRACK(S),
000108 190002**OPER**âARC0401I LARGEST EXTENTS FOR HSMM07 ARE CYLINDERS    533
000108 190003**OPER**$ARC0400I VOLUME HSMM08 IS 36% FREE, 001901 FREE TRACK(S),
000108 190003**OPER**âARC0401I LARGEST EXTENTS FOR HSMM08 ARE CYLINDERS    306
```

Figure 47: View Log Messages For Primary Space Detail Option

Since you are in ISPF, all the normal 'Find' and 'Scroll' commands are available to you. At the top of the display is the time that space management started working on its first volume. Max down to the bottom of the data set to see the time that space management ended.

I'm sure you can think of several reasons why you would need to find these messages quickly and easily. You can use this option to ensure that space management completes successfully, for example, to ensure space management completes within the panel provided in HSM's PARMLIB.

Press **PF3** until you return to the *HSM Reporter Master Panel*.

3.9 Tape Usage Summary - Master Panel Option 8

If you need to look at all of the tape activity for the reporting period (24 hours), we provide an option for that too. Key **8** in the 'Command' field on the *HSM Manager Master Panel* and press **Enter**. The *HSM Manager Tape Usage Summary* panel is displayed.

The following is an example of the *HSM Manager Tape Usage Summary* panel.

```
Row 1 to 18 of 393
 <V6 R1.xx>      HSM MANAGER TAPE USAGE SUMMARY
 OPTION ===>                                          SCROLL===> PAGE
  ENTER SORT FOR SORT OPTIONS. ENTER S BESIDE A VOLUME TO DISPLAY CONTENTS
  ENTER P FOR A BATCH PRINT OF THIS INFORMATION.
 S   VOLSER  TYPE    FUNCTION  COUNT
     ******   NF     BACK-UP   00025
     HSMM03   NF     BACK-UP   00205
     HSMM04   NF     BACK-UP   00003
     HSMM06   NF     BACK-UP   00075
     HSMM07   NF     BACK-UP   00012
     HSMM08   NF     BACK-UP   00079
     HSMM09   NF     BACK-UP   00039
     HSMM10   NF     BACK-UP   00060
     HSMM11   NF     BACK-UP   00020
     HSMM12   NF     BACK-UP   00008
     HSMM13   NF     BACK-UP   00051
     HSMM14   NF     BACK-UP   00031
     HSMM15   NF     BACK-UP   00182
     HSMM16   NF     BACK-UP   00042
     HSMM17   NF     BACK-UP   00023
     HSMM18   NF     BACK-UP   00005
     HSMM19   NF     BACK-UP   00150
     HSMM20   NF     BACK-UP   00012
```

Figure 48: HSM Manager Tape Usage Summary Panel

What HSM tape activities occurred? How many tapes were recycled? How many tape mounts were there for recall or recycle? How many tapes were used for backup? The answers to all these questions are provided here. You can sort by VOLSER, TYPE, FUNCTION, and COUNT in ascending or descending order in this panel.

Key **SORT** in the 'Command field' on the *HSM Manager Tape Usage Summary* panel and press **Enter**. The *HSM Manager Tape Usage Sort Options* panel is displayed.

The following is an example of the *HSM Manager Tape Usage Sort Options* panel.

```
<V6 R1.xx>          HSM MANAGER TAPE USAGE SORT OPTIONS
COMMAND ===>
 ENTER NUMERICAL VALUES  A=ASCENDING
 FOR SORT ORDER BELOW:   D=DESCENDING

 VOLSER ======> ACT      A    Note: It isn't necessary to provide a
 TYPE=========> DSN           sort order number in every field.
 FUNCTION=====> RC
 COUNT =======> RES

Enter REFRESH or R  to Reset sort to defaults
```

Figure 49: HSM Manager Tape Usage Sort Options Panel

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Press **PF3** to exit out of the *HSM Manager Tape Usage Sort Options* panel and return to the *HSM Manager Tape Usage Summary* panel.

From the *HSM Manager Tape Usage Sort Options* panel you can display more information. Tab down to the 'Select' column next to any of the tape VOLSERs, key **S**, and press **Enter**.

For example, key **S** in the 'Select' column next to HSMM12 on the *HSM Manager Tape Usage Sort Options* panel, and the HSM MANAGER DSNAME(S) FOR VOL HSMM12 panel is displayed. The data set names, last reference date, expiration date, and VOLSER information is provided in this panel. Scroll right for the HSM alias data set name.

Press **PF3** until you return to the *HSM Reporter Master Panel*.

4 Reports (Plans) and CDS Information

HSM Reporter/Manager provides 'pre-defined' reports that you can view interactively using the panels or in printed format. These reports are called plans. You can modify the provided plans or create your own.

Plans are available from the following options on the *HSM Reporter Master Panel*. The plans for migration will be discussed. The procedure is similar for the other options.

Option #	Option Name
2	MIGRATION/BACKUP OPTIONS
3	RECALL/RECOVER OPTIONS
4	LOG MESSAGE SUMMARY
C1	MCDS INFORMATION
C2	BCDS INFORMATION
C3	OCDS INFORMATION
C5	VOLUME DUMP INFORMATION

4.1 Migration/Backup Options - Master Panel Option 2

First we will look at the plans for MIGRATION/BACKUP OPTIONS.

Key **2** in the 'Command' field on the *HSM Reporter Master Panel* and press **Enter**. The *HSM Reporter Online Logfile Migration/Backup Options* panel is displayed.

The following is an example of the *HSM Reporter Online Logfile Migration/Backup Options* panel.

```
<V6 R1.xx>     HSM REPORTER ONLINE LOG FILE MIGRATION/BACKUP OPTIONS

 COMMAND==>

              1  MIGRATION RETURN CODE SUMMARY

              2  ML1 TO ML2 RETURN CODE SUMMARY

              3  BACKUP RETURN CODE SUMMARY

              4  FILTER OPTIONS FOR MIGRATE/BACKUP

              5  DISPLAY MIGRATION/BACKUP PLANS

              6  DISPLAY MIGRATION/BACKUP FILTERED FLAT FILES


  ENTER  END  TO TERMINATE ONLINE FUNCTION
```

Figure 50: HSM Manager Tape Usage Sort Options Panel

We want to look at **option 5** (DISPLAY MIGRATION/BACKUP PLANS). Key **5** in the Command field on the *HSM Reporter Online Logfile Migration/Backup Options* and press **Enter**. The *HSM Manager LBAK Plan Member List* panel is displayed.

The following is an example of the *HSM Manager LBAK Plan Member List* panel.

```
Row 1 to 17 of 20
 <V6 R1.xx>  HSM MANAGER LBAK PLAN MEMBER LIST
 OPTION ===>                                                  SCROLL==> PAGE
  SPECIFY SORT KEY==>        MLIST, TITLE, CREDT OR UPDT
  SPECIFY SORT TYPE=>        A FOR ASCENDING OR D FOR DESCENDING
  YOU MAY BROWSE (B), EDIT (E), VIEW (V) PRINT (P), FLAT (F) OR SUM (IF MSGS)
 COMMAND  MEMBER   ACTION             TITLE
          LBAK
          LBAK@LBL            FIELDS FOR MIGRATION/BACKUP
          LBAK001             FILTER NON ZERO MIGRATION RETURN CODES
          LBAK002             HM02 MIGRATION FROM PRIMARY BY AGE
          LBAK003             HM03 MIGRATION ERRORS BY RETURN CODE
          LBAK004             HM04 MIGRATION RETURN CODE 0 BY DSNAME
          LBAK005             HM05 MIGRATION RETURN CODE 0 BY PRIVOL/DSN
          LBAK0095            HM95 CORRECT NOT CATALOGED PROBLEM
          LBAK041             HM41 BACKUP ERRORS BY RETURN CODE
          LBAK042             HM42 BACKUP ERRORS BY AGE
          LBAK043             HM43 BACKUP DETAIL RETURN CODE 0 BY DSNAME
          LBAK044             HM44 BACKUP DETAIL RETURN CODE 0 BY AGE
          LBAK070             HM70 MIGRATE ACTIVITY BY PRIMARY VOLUME/DSN
          LBAK071             HM71 MIGRATE ACTIVITY BY PRIMARY VOLUME/TRACKS
          LBAK074             HM74 BACKUP ACTIVITY BY PRIMARY VOLUME/DSN
          LBAK075             HM75 BACKUP ERRORS BY PRIMARY VOLUME
          LBAK080             HM80 MIGRATIONS FOR EXTENT REDUCTION BY DSN, RETUR
```

Figure 51: HSM Manager LBAK Plan Member List Panel

As you can see from looking at the *HSM Manager LBAK Plan Member List* panel, there are several 'pre-defined' plans available for your immediate use. We will look at the reports from the plan LBAK002 (HM02 MIGRATION FROM PRIMARY BY AGE). Tab down next to the plan name, key **V** in the 'Command' field and press **Enter**. A pop-up panel will display the number of records read and the records selected. This count increases until all records have been read. The *HM02 MIGRATION FROM PRIMARY BY AGE* report panel is then displayed.

The following is an example of the *HM02 MIGRATION FROM PRIMARY BY AGE* report panel. Keep in mind that the data in your display is going to look different than ours.

```
<V6 R1.xx>  HM02 MIGRATION FROM PRIMARY BY AGE                  Row 1 of 3274
 OPTION ===>                                                   SCROLL==> PAGE
  ENTER O ON AT OPTION FOR LIST OF OPTIONS.
  PANEL 1 OF 3. SCROLL RIGHT FOR MORE INFORMATION.
  KEY S BESIDE ENTRY FOR RETURN/REASON CODES
 S   DSN                                          ACTION   RC RSNC  TRKS  AGE
    EXE.LANT.SALING.P20.MASTER1.G0646V00          MIGRATE  37 00000 26400   29
    SYS3.KING.DOME.NOT.HERE.G0001V00              MIGRATE  37 00000 89970   15
    EXE.LANT.SALING.PP.P20.MASTER1.G0484V00       MIGRATE  37 00000 26715   16
    EXE.QUANTUM.LEAP.P20.MASTER1.G0001V00         MIGRATE  37 00000 34755   29
    EXE.TRATER.STREAL.P20.MASTER1.G1187V00        MIGRATE  37 00000 25965   20
    EXE.LANT.SALING.P20.MASTER1.G0653V00          MIGRATE  37 00000 25965   22
    EXE.LANT.SALING.P20.MASTER1.G0675V00          MIGRATE  37 00000 27975   12
    SYS3.KING.DOME.ALL.GONE.G0002V00              MIGRATE  37 00000 88290   15
    EXE.TRATER.STREAL.P20.MASTER1.G1186V00        MIGRATE  37 00000 25965   20
    EXE.LANT.SALING.PP.P40OUT1.UNSORTED.G0246V00 MIGRATE  37 00000 19605   15
    EXE.LANT.SALING.PP.P20.MASTER1.G0492V00       MIGRATE  37 00000 27690   14
    EXE.TRATER.STREAL.P20.MASTER1.G1193V00        MIGRATE  37 00000 25965   17
    EXE.LANT.SALING.P20.MASTER1.G0664V00          MIGRATE  37 00000 26520   17
    EXE.LANT.SALING.P20.MASTER1.G0680V00          MIGRATE  37 00000 38805    8
    EXE.LANT.SALING.PP.P20.MASTER1.G0498V00       MIGRATE  37 00000 28740    9
    EXE.LANT.SALING.P20.MASTER1.G0678V00          MIGRATE  37 00000 28050    9
    EXE.LANT.SALING.P20.MASTER1.G0680V00          MIGRATE  37 00000 38805    8
    EXE.LANT.SALING.PP.P20.MASTER1.G0498V00       MIGRATE  37 00000 28740    9
```

Figure 52: HM02 MIGRATION FROM PRIMARY BY AGE Report Panel

You can manipulate the view by keying **O** in the 'OPTION' field and pressing **Enter**. The *Options* pop-up panel is displayed.

```
Enter Option number
 OPTION==>
   1 SORT
   2 FILTER
   3 PRINT
   4 SAVE
   F FORMAT AND SAVE
   5 PLAN
   6 TOTAL
   7 TAG    (manager)
   8 UNTAG  (manager)
   9 RUN    (manager)
  10 BATCH  (manager)
```

Figure 53: Options Pop-Up Panel

You have seen this *Options* pop-up panel before, so you are already familiar with how the options work. Now we will look at some reports from the MCDS.

Press **PF3** until you return to the *HSM Reporter Master Panel*.

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4.2 MCDS Information - Master Panel Option C1

Key **C1** In the 'Command' field on the *HSM Manager Master Panel* and press **Enter**. The *HSM Manager MCDS Format Options* panel is displayed.

The following is an example of the *HSM Manager MCDS Format Options* panel.

```
<V6 R1.xx>     HSM MANAGER MCDS FORMAT OPTIONS

 COMMAND==>


        1  FORMAT MCDS EXTRACT FILE ONLINE (TSO)
        2  SUBMIT BATCH JOB TO FORMAT MCDS EXTRACT FILE
        3  USE EXISTING REFORMATTED MCDS EXTRACT FILE
        4  ESTIMATE EXTRACTION FILE SIZE FOR MCDS

  enter  END  To terminate online function
```

Figure 54: HSM Manager MCDS Format Options Panel

This panel looks a little different than some of the others we have seen. To look at the current data in the MCDS, we need to run a FORMAT job first. You can do this interactively using **option 1,** or in batch using **option 2** on the *HSM Manager MCDS Format Options* panel. If you have a bunch of stuff migrated, it's a good idea to use **option 2** so you don't tie up your TSO session for a long wait.

Since we are on a system with a smaller number of records in the MCDS, we can request an online FORMAT. Key **1** in the 'Command field' on the *HSM Manager MCDS Format Options* panel and press **Enter**. A pop-up panel appears showing a progress display of the record count until it's all done.

```
HSM CONTROL DATA SET EXTRACTOR
  PROGRESS:

  RECORDS READ
  00013000
```

Once the FORMAT is completed, the *HSM Reporter Online MCDS Information* panel is displayed.

The following is an example of the *HSM Reporter Online MCDS Information* panel.

```
<V6 R1.xx>     HSM REPORTER ONLINE MCDS INFORMATION

 COMMAND==>


        1  MIGRATED DATA SETS BY SMS TYPE AND CLASSES
        2  MIGRATED DATA SETS BY MIGRATION VOLUME
        3  DATA SET NAME SEARCH
        4  FILTER MCDS DETAIL INFORMATION
        5  DISPLAY THE PLANS FOR THE MCDS
        6  DISPLAY FILTERED FLAT FILES FOR MCDS

  enter  END  To terminate online function
```

Figure 55: HSM Reporter Online MCDS Information Panel

4.2.1 MCDS Information Option 1 - Migrated Data Sets by SMS Type and Classes

Great! Now we are ready to look at the data. Key **1** in the 'Command' field on the *HSM Manager MCDS Format Options* panel and press **Enter**. The *HSM Manager MCDS SMS Summary* panel is displayed.

The following is an example of the *HSM Manager MCDS SMS Summary* panel.

```
Row 1 to 18 of 70
 <V6 R1.xx>       HSM MANAGER MCDS SMS SUMMARY
 OPTION ===>                                                SCROLL==> PAGE
  ENTER S IN THE SELECTABLE FIELDS FOR MORE DETAIL ON MANAGED DATA
                   CLASS
 S SMS CLASS       NAME        OCCURRENCES
    DATA           DCGDS       0003718
    DATA           DCLARGE     0001035
    DATA           DCLARG02    0000032
    DATA           DCLARG03    0000013
    DATA           DCLARG04    0000011
    DATA           DCMEDIUM    0000243
    DATA           DCNOORG     0170428
    DATA           DCPDS       0003077
    DATA           DCPS        0013478
    DATA           DCSMALL     0000531
    DATA           DCVS        0133770
    DATA           DCVSMALL    0000156
    DATA           NULL        0024119
    MANAGEMENT     MCBACKUP    0013818
    MANAGEMENT     MCBATCH     0021572
    MANAGEMENT     MCCMSA00    0000468
    MANAGEMENT     MCCMSA01    0001468
    MANAGEMENT     MCCSSBTH    0000711
```

Figure 56: HSM Manager MCDS SMS Summary Panel

The *HSM Manager MCDS SMS Summary* panel has all of the data classes, management classes, and storage classes with migrated data displayed. You may need to scroll down **(PF8)** to see the entire list. In the 'OCCURRENCES' column is the number of data sets migrated for each SMS class. This is a great way to see migration records by an individual Management Class for example.

To look at the data sets migrated by class, tab down next to one of the classes, key **S** in the 'Select' column, and press **Enter**. The *HSM Manager MCDS Display Of SMS Management Class* panel is displayed.

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The following is an example of the *HSM Manager MCDS Display Of SMS Management Class* panel.

```
Row 1 to 17 of 21,57
<V6 R1.xx>  HSM MANAGER MCDS DISPLAY OF SMS MANAGEMENT CLASS  MANAGEMENT
OPTION ===>                                                  SCROLL==> PAGE
 ENTER O IN OPTION FIELD FOR OPTIONS
 PANEL 1 OF 4. SCROLL RIGHT FOR MORE INFORMATION.   CURR DATE= 2000/111
                                          MIGRT- CREATE  REF     MIG
C           DSNAME                          ED  DATE    DATE    DATE
   CAI.PP.PANV.SMPCSI.CSI                    Y   1999256 1999336 1999340
   CICS410.EBISSG.BKUP.D000217.DFHCSD        Y   2000048 2000048 2000052
   CICS410.SPG.CICS.DFHDMPA                  Y   1999343 1999343 1999347
   CICS410.SPG.CICS.DFHDMPB                  Y   1999343 1999343 1999347
   CICS410.SPG.CICS.DFHGCD                   Y   1999343 1999343 1999347
   CICS410.SPG.CICS.DFHINTRA                 Y   1999343 1999343 1999360
   CICS410.SPG.CICS.DFHJACD                  Y   1999343 1999343 1999347
   CICS410.SPG.CICS.DFHJ01A                  Y   1999343 1999343 1999347
   CICS410.SPG.CICS.DFHJ01B                  Y   1999343 1999343 1999347
   CICS410.SPG.CICS.DFHJ01X                  Y   1999343 1999343 1999347
   CICS410.SPG.CICS.DFHLCD                   Y   1999343 1999343 1999347
   CICS410.SPG.CICS.DFHRSD                   Y   1999343 1999343 1999360
   CICS410.SPG.CICS.DFHTEMP                  Y   1999343 1999343 1999360
   CICS410.SPG.TEST10.DFHDMPA                Y   1999026 1999277 1999281
   CICS410.SPG.TEST10.DFHDMPB                Y   1999026 2000013 2000017
   CICS410.SPG.TEST10.DFHGCD                 Y   1999026 1999277 1999281
   CICS410.SPG.TEST10.DFHINTRA               Y   1999026 1999277 1999281
```

Figure 57: HSM Manager MCDS Display Of SMS Management Class Panel

The data sets are sorted alphabetically by default, but you can change this. Key O in the 'OPTIONS' field to display the *Options* pop-up panel. Use the **SORT** option in the *Options* pop-up panel to change how to sort the data sets. All of the other options in the *Options* pop-up panel are also available here.

Scrolling right shows more information about each data set.

```
XPIRE   LAST    MIG      ML0
DATE    BKUP    VOL      VOL
0000000 0000000 PL1032 PGL012
0000000 0000000 PL1026 PGL014
0000000 0000000 PL1038 PGL083
0000000 0000000 PL1021 PGL013
0000000 0000000 PL1021 PGL056
0000000 0000000 PL1027 PGL054
0000000 0000000 016785 PGS020
0000000 0000000 023334 PGS002
0000000 0000000 024093 PGS007
```

Figure 58: HSM Manager MCDS Display Of SMS Management Class Panel - scrolled right one time

Scroll right again for more great information.

```
MIG   T  M   N    ML0       MIGRATED
TIMES H  L   V    TRKS      KB2 BYTES
   1  9  1   N   18720     196958
   1  9  1   N   18720     196959
   2  8  1   N    7905      74646
   1  9  1   N    7830      74996
   1  8  1   N    1980      18406
   1  8  1   N    2520      23799
   1  3  2   N       1         10
```

Figure 59: HSM Manager MCDS Display Of SMS Management Class Panel - scrolled right two times

We like the fact that you can see the total number of tracks occupied on primary (ML0) and the KB2 bytes occupied on migration volumes. The 'TH' field represents a thrashing index computed from the age of the record divided by the number of times the data set has migrated. The 'NV' field shows 'V' if it's VSAM or 'NV' if it's a nonVSAM data set.

Scroll right again for more details.

```
DATA         MGMT     STORAGE SD
 CLASS        CLASS    CLASS   SP
 DCNOORG      GDGL     GDGL    N
 DCNOORG      GDGL     GDGL    N
 DCNOORG      GDGL     GDGL    N
 DCNOORG      GDGL     GDGL    N
 NULL         GDGL     GDGL    N
 NULL         GDGL     GDGL    N
```

Figure 60: HSM Manager MCDS Display Of SMS Management Class Panel - scrolled right three times

The 'SDSP' column shows you if the data set is in an small data set packing data set (SDSP) or not.

Scroll left until return to the beginning view of *HSM Manager MCDS Display Of SMS Management Class* panel.

Now pretend that you needed to recall a bunch of these data sets for some special project. You can select the data sets, tag them, and place them in a batch job right from the *HSM Manager MCDS Display Of SMS Management Class* panels.

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4.2.2 MCDS Information Option 2 - Migrated Data Sets by Migration Volume

Option 2 on the *HSM Reporter Online MCDS Information* panel shows which data sets are on either DASD or tape volumes. Key **2** in the 'Command' field and press **Enter**. The *HSM Reporter MCDS MIGVOL Summary Panel* will be displayed.

The following is an example of the *HSM Reporter MCDS MIGVOL Summary Panel*.

```
<V6 R1.xx>      HSM REPORTER MCDS MIGVOL SUMMARY PANEL
 OPTION ===>                                    SCROLL==> CSR
  ENTER S IN THE SELECTABLE FIELD FOR  MORE DETAIL
              UNIT
 S  VOLUME    TYPE
    HM0001    3480
    HM0002    3480
    HM0003    3480
    HM0004    3480
    HM0005    3480
    HM0006    3480
    HM0007    3480
    HM0008    3480
    HM0009    3480
    HM0010    3480
    HM0011    3480
    HM0012    3480
    HM0013    3480
    HM0014    3480
    HM0015    3480
    HM0016    3480
    HM0017    3480
    HM0018    3480
```

Figure 61: HSM Reporter MCDS MIGVOL Summary Panel

This option comes in *real handy* when the tape SILO eats an ML2 tape and you need to delete all the migration records and recover all of the data sets from backup. You can do all of those functions right here in this panel!

4.2.3 MCDS Information Option 3 - Data Set Name Search

Option 3 on the *HSM Reporter Online MCDS Information* panel allows you to key in a data set name or a data set name mask. Key **3** in the 'Command' field and press **Enter**. The *HSM Reporter Online MCDS DSNAME Search Panel* will be displayed.

The following is an example of the *HSM Reporter Online MCDS DSNAME Search Panel*.

```
<V6 R1.xx>   HSM REPORTER ONLINE MCDS DSNAME SEARCH PANEL

 OPTION ===>
              SPECIFY ACS FILTER PATTERN
  DSNAME==> USERHLQ.*.ASM


  Enter END or PFK3 to terminate this panel
```

Figure 62: HSM Reporter Online MCDS DSNAME Search Panel

For example, your co-worker has just returned from a leave of absence and you need to recall all of his TSO utility data sets. You can key a partial data set name, like 'XX00056.**' and get an entire list of the data sets in migration. You can then tag and place them in a batch job with the HSM command 'HRECALL'.

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4.2.4 MCDS Information Option 4 - Filter MCDS Detail Information

Key **4** in the 'Command' field on the *HSM Reporter Online MCDS Information* panel and press **Enter**. The *HSM Manager MCDS SMS Report Filter Options* panel will be displayed.

The following is an example of the *HSM Manager MCDS SMS Report Filter Options* panel.

```
<V6 R1.xx>       HSM MANAGER MCDS SMS REPORT FILTER OPTIONS
 COMMAND ===>
    FILTER OPTION==>         enter F to perform filter in TSO BATCH
  LABEL          OPCODE    ARGUMENT
  DSNAME      =>         =>
  MIGRATED    =>         =>
  SDSP        =>         =>
  THRASHING   =>         =>
  CREATE DATE =>         =>
  REF DATE    =>         =>
  MIG DATE    =>         =>
  LAST BKUP   =>         =>
  XPIRE DATE  =>         =>
  MIG TIMES   =>         =>
  MIGVOL      => EQ      => 001512
  ML0VOL      =>         =>
  ML          =>         =>
  NV          =>         =>
  ML0TRKS     =>         =>
  KB2         =>         =>
  DATA CLASS  =>         =>
  MGNT CLASS  =>         =>
  STOR CLASS  =>         =>
 Enter REFRESH or R  to Reset filter to defaults
```

Figure 63: HSM Manager MCDS SMS Report Filter Options Panel

From this panel you can filter data sets by any or a combination of the options. For example, you want to see all data sets that were created before '1999001' and are still on ML0 (primary DASD). You can get a list to match that criteria by keying in the filters and pressing **Enter**. Press **PF1** in this panel to get help with the OPCODES or refer to the *HSM Reporter/Manager User Guide* (document 002-0202) for more information about these options.

4.2.5 MCDS Information Option 5 - Display the Plans for the MCDS

Now back to plans. Key **5** in the 'Command' field on the *HSM Reporter Online MCDS Information* panel and press **Enter**. The *HSM Manager MCDS Plan Member List* panel is displayed.

The following is an example of the *HSM Manager MCDS Plan Member List* panel.

```
Row 1 to 14 of 14
 <V6 R1.xx>  HSM MANAGER MCDS PLAN MEMBER LIST
 OPTION ===>                                                  SCROLL==> PAGE
  SPECIFY SORT KEY==>        MLIST, TITLE, CREDT OR UPDT
  SPECIFY SORT TYPE=>        A FOR ASCENDING OR D FOR DESCENDING
  YOU MAY BROWSE (B), EDIT (E), VIEW (V) PRINT (P), FLAT (F) OR SUM (IF MSGS)
 COMMAND  MEMBER   ACTION            TITLE
          MCDS@LBL            FIELDS FOR MCDS PLANS
          MCDSMCDS
          MCDSSMS             SPECIAL PLAN TO PRINT THE SMS SUMMARY INFO
          MCDS005             LIST ALL DATA SETS REFERENCED LAST YEAR
          MCDS009             DISPLAY THRASHING DSNAMES FOR LAST SEVEN DAYS
          MCDS010             LIST ALL SPECIFY DATA SETS DIDN'T REF BY SPECIFY P
          MCDS011             DISPLAY DATA SETS MIGRATED AND RECALLED THIS WEEK
          MCDS018             PRINT MIGRATION THRASHING FOR LAST 14 DAYS
          MCDS021             PRINT DATA SETS TO EXPIRE IN 100 DAYS
          MCDS101             DISPLAY MIGRATED DATA SETS 2 YEARS OLD
          MCDS102             HRO102 CURRENT WEEK MIGRATION THRASHING REPORT
          MCDS120             DISPLAY DSN(S)W/RECALL COUNT GT 30
          MCDS200             DATA SETS IN MIGRATION SYSTEM SOFTWRE
          MCDS202             SYSTEM SOFT DS IN MIGRATION
 ****************************** Bottom of data ********************************
```

Figure 64: HSM Manager MCDS Plan Member List Panel

Just like in the other plans panels, a few 'pre-defined' reports are provided for your use, or you can modify or make your own plans.

The first plan listed on the *HSM Manager MCDS Plan Member List* panel is 'MCDS@LBL'. This is a member that lists all of the filter labels you can use to create your own plan.

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The following is an example of the filter label listing from plan MCDS@LBL.

```
000001 FIELDS FOR MCDS PLANS
000002 THIS IS NOT A PLAN
000003 FILTER LABELS FOR PLAN ARGUMENTS FOR MCDS FOLLOW
000004 ASDSN    EQ AAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAA AND
000005 ASCRDT   EQ 9999999      (CREATION DATE YYYYDDD)          AND
000006 ASRFDT   EQ 9999999      (LAST REFERENCED DATE YYYYDDD)   AND
000007 ASMGDT   EQ 9999999      (MIGRATION DATE YYYYDDD)         AND
000008 ASBKDT   EQ 9999999      (BACK-UP DATE YYYYDDD)           AND
000009 ASXPDT   EQ 9999999      (EXPIRATION DATE YYYYDDD)        AND
000010 ASRCNT   EQ 99999        (RECALL COUNT)                   AND
000011 ASVOL    EQ AAAAAA       (MIGRATION VOLUME)               AND
000012 ASL0     EQ AAAAAA       (PRIMARY VOLUME MIGRATED FROM)   AND
000013 ASLVL    EQ 9            (LEVEL OF MIGRATION 1 OR 2)      AND
000014 ASVNV    EQ A            (VSAM/NONVSAM INDICATOR V/N)     AND
000015 ASTRKS   EQ 999999       (NUMBER OF TRACKS MIGRATED)      AND
000016 ASKB2    EQ 9999999999   (NUMBER OF 2KB BLOCKS MIGRATED)  AND
000017 ASDCLSS  EQ AAAAAAAA     (DATA CLASS NAME)                AND
000018 ASMCLSS  EQ AAAAAAAA     (MANAGEMENT CLASS NAME)          AND
000019 ASSCLSS  EQ AAAAAAAA     (STORAGE CLASS NAME)             AND
000020 ASMG     EQ A            (CURRENTLY MIGRATED Y/N)         AND
000021 ASDP     EQ A            (IN SDSP Y/N)                    AND
000022 AI       EQ 9            (THRASHING INDEX 1-9)            AND
 ****** ***************************** Bottom of Data ***************************
```

Figure 65: List Of Filter Labels In Member MCDS@LBL

The following example of plans MCDS005 is provided so you can see how to use the fields.

```
000001 LIST ALL DATA SETS REFERENCED LAST YEAR
 000002 SORT FOLLOWS
 000003 SMG,C,A
 000004 FILTER FOLLOWS
 000005 ASRFDT   LT *-365                                          AND
 000006 ASRFDT   GT *-730                                          AND
 000007 ASMG     EQ Y                                              AND
 ****** ***************************** Bottom of Data ******************
```

Figure 66: MCDS005 Pre-Defined MCDS Plan

4.3 Create/Modify A Plan

Plans are a convenient way of customizing reports and extending the filtering capabilities of HSM Reporter/Manager. Plans capture the most current sort and filter selections for the Migration data.

4.3.1 Create A New Plan

Before you create a plan make sure the desired sort and filter values have been set.

Plans may be invoked from the plans option (**option 5**) of the *HSM Reporter Online MCDS Information* panel.

The *Build Plan For MCDS* panel will be displayed.

```
<V6 R1.xx> HSM REPORTER ONLINE BUILD PLAN FOR MCDS
 COMMAND ===>
            PLAN TYPE==>  MCDS
  SPECIFY PLAN NUMBER==>  207   (101 through 9999)
  EDIT, BROWSE or RETURN  EDIT
   Specify description of plan below:
  Title==> DEMONSTRATE PLAN

   Note:  001       through 100       is reserved for system plans
   specify user plan MCDS101 to MCDS9999
```

In the above panel you must specify:

- A plan number at the SPECIFY PLAN NUMBER prompt.
- The action to be performed at the completion of the plan build (EDIT, BROWSE or RETURN).
- A Title. The title is important because it will be displayed on the plan's selection panel and serves as an ‘eye-catcher’ for what the plan contains. Be sure to enter something meaningful here.

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4.3.2 Modify An Existing Plan

Another way to create a plan is to copy an existing plan and modify it to suit your needs. You can do this easily by editing one of the plans and creating a new member. .

Tab down next to one of the 'pre-defined' plans, key **E** in the 'Command' field and press **Enter**. For this example we are going to use the plan displayed in the previous topic.

Key in **CREATE MCDS9999** (plans for the MCDS need to start with MCDS), then tab down to line '000001' and key **C99999** over '000001'. Press **Enter**. In the upper right hand corner of your screen, you will see the words 'MEMBER MCDS9999 CREATED'. You can immediately edit this plan by keying **EDIT MCDS9999** in the 'Command' field, and pressing **Enter**.

Now you are in ISPF Edit for member MCDS9999:

```
EDIT       SYS3.MAINSTAR.HRP5201.PLANS(MCDS9999) - 01.03   Columns 00001 00072
 Command ===>                                                 Scroll ===> CSR
 ****** ***************************** Top of Data ******************************
 000001 LIST ALL DATA SETS REFERENCED LAST YEAR
 000002 SORT FOLLOWS
 000003 SMG,C,A
 000004 FILTER FOLLOWS
 000005 ASRFDT   LT *-365                                        AND
 000006 ASRFDT   GT *-730                                        AND
 000007 ASMG     EQ Y                                            AND
 000008 ASDSN    AN HSMACT.**                                    AND
 ****** *************************** Bottom of Data ***************************
```

Figure 67: Use ISPF Edit To Create A New Plan By Modifying A Pre-Defined Plan

Now we will change this plan to exclude all data sets that begin with 'HSMACT'. We repeated the last line and added statement '000008'.

Note that all of the OPCODES we saw before work here as well. They are EQ (equal), NE (not equal), LT (less than), GT (greater than), GE (greater than or equal to), LE (less than or equal to), LK (like), NK (not like), AE (ACS equal), or AN (ACS not equal). We used AN (ACS not equal) to filter out those data sets that matched the partial data set names 'HSMACT.**'. If we were to use 'HSMACT.**', HSM Reporter/Manager would look for any data set names beginning with 'HSMACT.**' to filter out and would not find any data set names that match.

4.4 Execute Your Plan

To execute the plan you created, press **PF3** to exit out of the *HSM Reporter Online MCDS Information* panel. This is because we created a new member and, just like a PDS, you need to back out and return to display the new member.

Key **5** in the 'Command' field again to display the plans. The new plan is shown in the *HSM Manager MCDS Plan Member List* panel.

The following is an example of the *HSM Manager MCDS Plan Member List* panel.

```
Row 1 of 16
<V6 R1.xx>  HSM MANAGER MCDS PLAN MEMBER LIST
OPTION ===>                                                  SCROLL==> PAGE
 SPECIFY SORT KEY==>        MLIST, TITLE, CREDT OR UPDT
 SPECIFY SORT TYPE=>        A FOR ASCENDING OR D FOR DESCENDING
 YOU MAY BROWSE (B), EDIT (E), VIEW (V) PRINT (P), FLAT (F) OR SUM (IF MSGS)
COMMAND  MEMBER   ACTION            TITLE
         MCDS@LBL             FIELDS FOR MCDS PLANS
         MCDSML2A             STANDARD DATA SETS IN ML2 GROUP A
         MCDSML2B             STANDARD DATA SETS IN ML2 GROUP B
         MCDSML2C             STANDARD DATA SETS IN ML2 GROUP C
         MCDSML2D             STANDARD DATA SETS IN ML2 GROUP D
         MCDSML2E             STANDARD DATA SETS IN ML2 GROUP E
         MCDSSMS              SPECIAL PLAN TO PRINT THE SMS SUMMARY INFO
         MCDS005              LIST ALL DATA SETS REFERENCED LAST YEAR
         MCDS009              DISPLAY THRASHING DSNAMES FOR LAST SEVEN DAYS
         MCDS011              DISPLAY DATA SETS MIGRATED AND RECALLED THIS WEEK
         MCDS018              PRINT MIGRATION THRASHING FOR LAST 14 DAYS
         MCDS021              PRINT DATA SETS TO EXPIRE IN 100 DAYS
         MCDS101              DISPLAY MIGRATED DATA SETS 2 YEARS OLD
         MCDS102              HRO102 CURRENT WEEK MIGRATION THRASHING REPORT
         MCDS120              DISPLAY DSN(S)W/RECALL COUNT GT 30
   V     MCDS9999             LIST ALL DATA SETS REFERENCED LAST YEAR
******************************* Bottom of data ********************************
```

Figure 68: Display Your New Plan In The MCDS Plan Member List

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To execute the plan, tab down to the 'Command' field next to our new plan (MCDS9999), key **V**, and press **Enter**.

Your plan is executed and your report is displayed.

```
Row 1 of 17
<V6 R1.xx>  HSM MANAGER LIST ALL DATA SETS REFERENCED LAST YEAR
OPTION ===>                                                SCROLL==> PAGE
 ENTER O IN OPTION FIELD FOR OPTIONS
 PANEL 1 OF 4. SCROLL RIGHT FOR MORE INFORMATION.   CURR DATE= 2000/060
                                              MIGRT- CREATE  REF     MIG
C           DSNAME                              ED  DATE    DATE    DATE
   GEN.TSTG00.F.GLMASTR.SEQ                      Y   1999013 1999013 2000008
   GEN.TSTG00.GLMASTR.SEQ                        Y   1999013 1999013 2000008
   GEN.TSTG00.RPTPRT5                            Y   1999013 1999015 2000059
   GTO.HLS.TEST.CO600FH.WORK08                   Y   1998327 1998327 1999354
   GTO.HLS.TEST.CO600FH.WORK14                   Y   1998327 1998327 1999354
   GTO.HLS.TEST.CO600FH.WORK22                   Y   1998327 1998327 1999354
   GTO.HLS.TEST.LRSMW                            Y   1998342 1998343 1999354
   GTO.REL3.HISTY200.CO253.TEST                  Y   1998329 1998329 1999354
   NC99102.SRCHFOR.LIST                          Y   1998297 1998298 1999342
   NC99312.LC.CRED.CARD                          Y   1999018 1999018 1999354
   NC99905.CM.VLF1                               Y   1998279 1998279 1999344
   NC99312.LC.CRED.CARD.ORDERS                   Y   1999008 1999018 1999354
   NC99404.LIST.SUPERC.CYBAR                     Y   1998210 1998258 1999354
   NC99404.LIST.SUPERC.EXPORT10                  Y   1998181 1998181 1999354
   NC99404.LIST.SUPERC.O6768                     Y   1998142 1998142 1999354
   NC99404.SUPERC.P9CNVT                         Y   1998125 1998140 1999354
   NC99527.SUPERC.STMT030                        Y   1998231 1998266 1999342
```

Figure 69: Display Of Plan Report

4.5 Print Your Plan

Of course, in addition to displaying your plan. Keying **O** in the 'OPTION' field of the *HSM Manager MCDS Plan Member List* panel and press **Enter**. The *Options* pop-up panel will be displayed. To print the plan, select **option 3** (Print).

The following is an example of the *Options* pop-up panel.

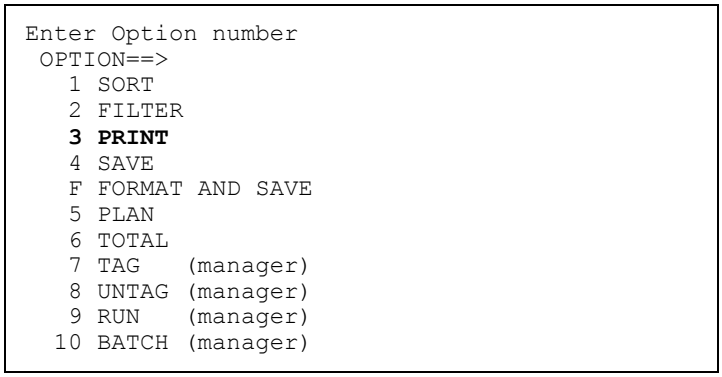


Figure 70: Use Options Pop-Up Panel To Print Your Report

After submitting your print job, press **PF3** to exit out of the *HSM Manager MCDS Plan Member List* panel.

5 HSM Health Reports and Automatic Charts

The HSM Reporter/Manager Health Reports were developed to inform the user what automatic scheduled functions of HSM were not completed and how long since last completed. There these reports, along with Automatic Management Charts, provide more extensive information. HSM errors are not the only errors in your HSM CDSs. Other types of errors could be thrashing, automatic scheduled tasks not completing or touching every volume, lost or damaged HSM ML2 or backup tapes, HSM tapes in failed create status, etc.. There are many products available on the market today that notify you if an HSM scheduled task did not finish successfully or provide some other *reactive* information. However, these products do not help you be *proactive* in your job.

The HSM Reporter/Manager Health Reports help you to be proactive. These reports read critical information gathered from your FSR records, MWE (manage work element) records, SMS constructs for volumes and storage groups, along with the MCD (mgmt control record). The volume statistic record will tell us about the volume activity for reporting. These new Health Reports put the product to a new directive of reporting not only on what did not get done, but also what was suppose to get done, and what did get done.

5.1 HSM Health Reports - Master Panel Option C4

Key **C4** (HSM Health Reports) in the 'Command' field of the *HSM Manager Master Panel* and press **Enter**. The *HSM Health Format Options* panel will be displayed.

The following is an example of the *HSM Health Format Options* panel.

```
   HSM MANAGER HSM HEALTH FORMAT OPTIONS
COMMAND==>
       1  SET HEALTH REPORT PARAMETERS
       2  SUBMIT JOB TO FORMAT HEALTH REPORTS
       3  ANALYZE HEALTH REPORT INTERACTIVELY
       4  RETRIEVE HISTORICAL REPORT TO ANALYZE
```

Figure 71: HSM Health Format Options Panel

5.1.1 HSM Health Format Option 1 - Set Health Report Parameters

First you will want to set up your defaults for the Health Reports.

Option 1 on the *HSM Health Format Options* panel is where you want to set a common default for notification of concern to set up your defaults. Key **1** in the 'Command' field and press **Enter**. The *HSM Health Report Settings* panel will be displayed.

The following is an example of the *HSM Health Report Settings* panel.

```
HSM MANAGER HSM HEALTH REPORT SETTINGS
COMMAND ===>
SPECIFY DAYS FOR EXCEPTION REPORTING===> 02
SPECIFY LIMIT FOR EXCEPTION REPORTING==> 10
HISTORY GDG BASE==> TL01.HSMHELTH.HISTORY
```

Figure 72: HSM Health Report Settings Panel

In the above panel we set the number of 'Days Exception Reporting' to '02'. This will only report the detailed HSM automatic functions that have not completed for two or more days. This number can be any number you choose. Be aware that the higher the number, the storage and data availability could be at risk.

The 'Limit For Exception Reporting' is set to '10'. It is normal for a volume to get migration errors during automatic functions (i.e., data set in use, no space on ML1 volume, unsupported data set, etc.); however if a volume has more then 10 reported errors per volume, you want to know about it. This setting can also be set to any number you choose.

The 'History GDG Base' will be what the 'extraction data' will be named. This is an excellent use for historical reporting and research. You can define a GDG base to the number of historical extracts you wish to keep.

After you have set your defaults, press **Enter** and then **PF3 (End)** to exit out of this panel. You will be back to the *HSM Health Format Options* panel.

5.1.2 HSM Health Format Option 2 - Submit Job To Format Health Reports

Next you will need to set your defaults for the reports and other functions.

Option 2 on the *HSM Health Format Options* panel will automatically build and the JCL needed to extract the required records and extraction file. Key **2** in the 'Command' field and press **Enter**.

The following is an example of the generated JCL. Note that STEP01 uses an authorized program so the STEPLIB must be an authorized library.

```
//DCDHSM    JOB (11111),'DAND',REGION=6M,
//   MSGCLASS=X,CLASS=A,NOTIFY=DCD,TIME=1
//*
//*
//STEP01 EXEC PGM=HASHMVT,REGION=4096K
//STEPLIB   DD  DSN=DCD.FAUDIT.LOAD,DISP=SHR
//MVTOUT    DD  DSN=&&MVT,DISP=(,PASS),
//          SPACE=(TRK,(5,5)),UNIT=SYSDA,
//          DCB=(RECFM=FB,LRECL=100,BLKSIZE=10000)
//*
//STEP02    EXEC PGM=HRO30000
//STEPLIB   DD  DSN=TL01.HRPvrmm.LOAD,DISP=SHR
//MVTOUT    DD  DSN=&&MVT,DISP=(OLD,DELETE)
//MCDS      DD  DSN=SIS.HSM.SPLIT.MCDS1,DISP=SHR
- - - - - - - - - - - - - - - - - - - - - - - - - - - - -
//STEP03    EXEC PGM=HRO32000,
//          PARM='2,2'
//STEPLIB   DD  DSN=TL01.HRPvrmm.LOAD,DISP=SHR
//DBHEALTH  DD  DSN=TL01.HSMHELTH.BASE,DISP=SHR
//HLTHSUM   DD  DSN=TL01.HSM.HEALTH.SUM,
//   DISP=SHR
//HLTHERRS  DD  DSN=TL01.HSM.HEALTH.ERRORS,
//   DISP=SHR
//DBSPOOL DD DSN=TL01.HSMHELTH.HISTORY(+1),
// UNIT=SYSDA,SPACE=(TRK,(25,10)),
// DCB=(RECFM=FB,LRECL=300,BLKSIZE=23400)
```

Figure 73: Sample Generated JCL

This JCL may be saved to a job library and scheduled on a regular, controlled basis, or submitted as is. In the above example, if the history base in the setup panel was left blank, then the DBSPOOL statement would generate DSN=NULLFILE.

> **Note:** You can also build the JCL from the *HSM Manager Master Panel*. Refer to the HSM Reporter/Manager User Guide for instructions on using **Option B** on the *HSM Manager Master Panel*.

5.1.3 HSM Health Format Option 3 - Analyze Health Report Interactively

Option 3 on the *HSM Health Format Options* panel is where the product will report the information it has found. Key **3** in the 'Command' field and press **Enter**. The *HSM Manager Health Panel* will be displayed.

The following is an example of the *HSM Manager Health Panel*.

```
HSM MANAGER HEALTH PANEL
COMMAND==>
           HSM HEALTH REPORT SELECTIONS
           1  SMS/HSM MOUNTED VOLUME REPORTS
           2  HSM MIGRATION REPORTS
           3  HSM BACK-UP REPORTS
           4  HSM VOLUME DUMP REPORTS
           5  STORAGE GROUP SUMMARY
```

Figure 74: HSM Manager Health Panel

We want to check that Primary and Secondary Space Management not only started and completed, but that it also touched all of your volumes. To do, this key **2** in the 'Command' field of the *HSM Manager Health Panel* and press **Enter** to display the HSM Migration Reports.

The following is an example of the HSM Migration Reports.

```
ENTER V TO DISPLAY ALL DETAIL ABOUT VOLUME OR B TO BROWSE ERROR RELATED DETAIL
  NUMBER   COUNT     MESSAGE
MIGR000     59     HSM MANAGED VOLUME MIGRATION CRITERIA HAS BEEN MET
MIGR001     10     VOLUME EXPECTING D.S.M. HAS NOT RUN IN 002 DAYS
MIGR002      7     VOLUME EXPECTS INTVL MIGR, NOT RUN IN 002 DAYS
MIGR003      3     VOLUME EXCEEDS HIGH OCCUPANCY THRESHOLD VALUE
MIGR004      3     MIGRATION ERRORS EXCEED 000010 FAILURES
MIGR005     18     VOLUME HAS HAD NO RECORDED VSR ACTIVITY IN 003 DAYS
MIGR006      0     SDSP IS DISABLED ON ML1 VOLUME
MIGR007      0     PERCENT ALLOCATED ON AN ML1 VOLUME EXCEEDS THRESHOLD
MIGR008     15     ML1 VOLUME HAS HAD NO RECORDED ACTIVITY IN 002 DAYS
MIGR009      2     ML1 VOLUME MIGRATION ERRORS EXCEED 000010 FAILURES
MIGR010      2     ML1 VOLUME RECALL ERRORS EXCEED 000010 FAILURES
```

Figure 75: Example of HSM Migration Reports

The HSM Migration Reports provide you with a quick view of the migration work performed since the extraction. From this report, you see where the '02' days and the '10' for exceptions you entered in previously (from **option 1** on the *HSM Health Format Options* panel) comes into the reporting.

Looking quickly you can see there are some concerns to be looked into. From the above panel, **59** volumes successfully met criteria, **10** volumes have not had DSM (data set management) performed in **10** or more days (the window may be too short), another **7** volumes have not had an Interval Migration done in **2** or more days, as well as a whole list of other errors. In addition, you see that several volumes have data sets that experienced some type of error during these functions.

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The Migration Report, MIGR001, reported volumes that have not had Data Set Management performed for two or more days. To get a listing of the volumes meeting this criteria, you want to drill down to the volume level. To do this simply put a 'V' to the left of the message (MIGR001).

The following is an example of the results of displaying all the detail about the volumes from the Migration Report MIGR001.

```
2003062SYS  STORAGE      DEVICE  S H M S E SPACE-  MANAGEMENT A  AUTO-BACKUP
VOLSER ID   GROUP        TYPE    M M V D N DATE    TIME       M  DATE    TIME
ARCHV1 DVLP ARCHIVE      3390-3  Y Y Y N N 2003062 00:12:26   Y  2000236 07:15:48
ARCHV2 DVLP ARCHIVE      3390-3  Y Y Y N N 2003062 00:12:52   Y  2000236 07:15:48
BOOK01 DVLP SGBOOKS      3390-3  Y Y Y N N 2003062 00:14:03   Y  1900001 00:00:00
CASE01 DVLP SGCASE       3390-3  Y Y Y N N 2003062 00:12:01   Y  1900001 00:00:00
CASE02 DVLP SGCASE       3390-3  Y Y Y N N 2003062 00:12:01   Y  1900001 00:00:00
CASE03 DVLP SGCASE       3390-3  Y Y Y N N 2003062 00:12:02   Y  1900001 00:00:00
CONCP0 DVLP CONCPY       3390-3  Y Y Y N N 2003062 00:13:43   Y  1900001 00:00:00
CONCP1 DVLP CONCPY       3390-3  Y Y Y N N 2003062 00:13:44   Y  1900001 00:00:00
CONCP2 DVLP CONCPY       3390-3  Y Y Y N N 2003062 00:13:44   Y  1900001 00:00:00
CONCP3 DVLP CONCPY       3390-3  Y Y Y N N 2003019 00:13:45   Y  1900001 00:00:00
```

Figure 76: Output from Migration Report MIGR001

In the above panel, if the current Julian date is 2003065, all the volumes listed have not had Data Set Management performed on them for at least three days. You can also see that one volume has been in error for 46 days. The HSM Health Reports will continue to report on all volumes in error until they are referenced and checked by HSM automatic functions. The process then starts over. You can use this same process to gather information and findings from the other Health Reports and messages.

A second critical Health Report you will want to check *every day* will be **option 3** (HSM Back-Up Reports) on the *HSM Manager Health Panel*. This information is gathered around your HSM Incremental backups and you will want to verify that the incremental backups were done for all volumes expecting to have backups done on changed data sets. *If this is not happening, and these reports are not checked, the risk for data loss is increased.*

For information about the HSM Health Reports, refer to *Chapter 6, HSM Health Reports*, in the HSM Reporter/Manager User Guide.

5.2 Automatic Charting And Trending Analysis

The HSM Health Reports also contains an automatic charting and trending analysis feature. These automatic charts use your PC to provide a quick visual look of what your HSM environment is doing, as well as a historical trend analysis. A macro is provided to access these charts and is shipped with the HSM Reporter/Manager product.

To create the special comma separated file, from your mainframe, follow the normal extraction process of **Option B** on the *HSM Manager Master Panel* and then **option 8**. **Option 8** will automatically create the MVS flat file you will need. You need to transfer this file to your PC every day. Once this file is uploaded to your PC, you will need to copy the macro into your Excel or Lotus Notes directory and then simply open it to review the charts.

5.2.1 Lists of Automatic HSM Health Reports Charts

Following are lists of all the different HSM Health Reports Charts that are automatically generated. By reviewing these charts, and the historical data provided, you will have a *total* understanding of your HSM environment.

Migration Charts

- Migrated Delete GB Read and Written
- Migrated Delete Q Time, Vol mount wait time, and Process time.
- Migrated Delete Total Age in Days of the Data Sets Processed
- Migrated Delete Requests Failing (RQF), User Requests (RQU), and System Requests (RQS)
- Migrated Delete Tracks Read and Written
- Migrated Delete DSNs

Recycle Charts

- Recycle GB Read and Written
- Recycle Q Time, Vol mount wait time, and Process time.
- Recycle Total Age in Days of the Data Sets Processed
- Recycle Requests Failing (RQF), User Requests (RQU), and System Requests (RQS)
- Recycle Tracks Read and Written
- Recycle DSNs Moved

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Primary to Level 1 Movement Charts

- Primary to L1 GB Read and Written
- Primary to L1 Q Time, Vol mount wait time, and Process time.
- Primary to L1 Total Age in Days of the Data Sets Processed
- Primary to L1 Requests Failing (RQF), User Requests (RQU), and System Requests (RQS)
- Primary to L1 Tracks Read and Written
- Primary to L1 DSNs Moved

Primary to Level 2 Movement Charts

- Primary to L2 GB Read and Written
- Primary to L2 Q Time, Vol mount wait time, and Process time.
- Primary to L2 Total Age in Days of the Data Sets Processed
- Primary to L2 Requests Failing (RQF), User Requests (RQU), and System Requests (RQS)
- Primary to L2 Tracks Read and Written
- Primary to L2 DSNs Moved

Level 1 to Level 2 Movement Charts

- L1 to L2 GB Read and Written
- L1 to L2 Q Time, Vol mount wait time, and Process time.
- L1 to L2 Total Age in Days of the Data Sets Processed
- L1 to L2 Requests Failing (RQF), User Requests (RQU), and System Requests (RQS)
- L1 to L2 Tracks Read and Written
- L1 to L2 DSNs Moved

Recall Charts

- L1 to Primary GB Read and Written
- L1 to Primary Q Time, Vol mount wait time, and Process time.
- L1 to Primary Total Age in Days of the Data Sets Processed
- L1 to Primary Requests Failing (RQF), User Requests (RQU), and System Requests (RQS)
- L1 to Primary Tracks Read and Written
- L1 to Primary DSNs Moved
- L2 to Primary GB Read and Written
- L2 to Primary Q Time, Vol mount wait time, and Process time.
- L2 to Primary Total Age in Days of the Data Sets Processed
- L2 to Primary Requests Failing (RQF), User Requests (RQU), and System Requests (RQS)
- L2 to Primary Tracks Read and Written
- L2 to Primary DSNs Moved

Backup Charts

- Backup Delete GB Read and Written
- Backup Delete Q Time, Vol mount wait time, and Process time
- Backup Delete Total Age in Days of the Data Sets Processed
- Backup Delete Requests Failing (RQF), User Requests (RQU), and System Requests (RQS)
- Backup Delete Tracks Read and Written
- Backup Delete DSNs Moved
- Backup GB Read and Written
- Backup Q Time, Vol mount wait time, and Process time.
- Backup Total Age in Days of the Data Sets Processed
- Backup Requests Failing (RQF), User Requests (RQU), and System Requests (RQS)
- Backup Tracks Read and Written
- Backup DSNs Moved

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HSM Overall Charts

- Data Set Restores Requested and Failed
- Volume Restores Requested and Failed
- System Requested Dump Copies, User Requested Dump Copies, and Failing Copies
- System Requested Volume Dumps, User Requested Volume Dumps, and Failing Dumps
- Tape Volumes Recycled and L2 Volumes Recycled
- Number of Volumes Backed Up
- Volumes Migrated and Number Migrated due to Lack of Space
- HSM Step Time for the HSM ASCB
- HSM Startups, ABENDs and Shutdowns

5.2.2 Examples of Automatic HSM Health Reports Charts

The following chart example reports the Primary to Level 1 Gigabytes Read and Written.

- **P1 GB Read** indicates how much data was being move for a specific day.
- **P1 REQ TIM** provides a visual for the total elapsed time it took to complete the task.

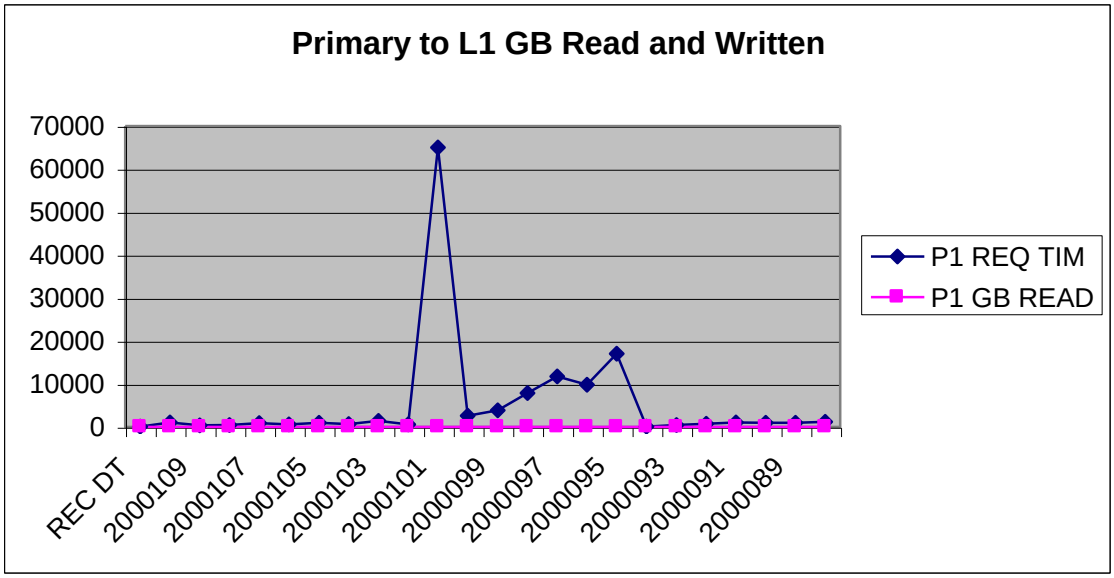


Figure 77: Health Report Chart - Primary to L1 GB Read and Written

When you see spikes (as shown in the above example), you can quickly check and see if there were any issues from the previous night's work. A good example could have been a month-end or quarter-end processing where a high amount of data was being recalled. This is also an excellent chart for monitoring the possibility of thrashing.

The following chart example reports on the Primary to L1 Q Time, Vol Mount Wait Time, and Process Time.

- **P1 VOLMTTM** indicates volume mount wait time.
- **P1 Q TIME** indicates queue time for requests.
- **P1DSA** indicates the total age in days of data sets processed.

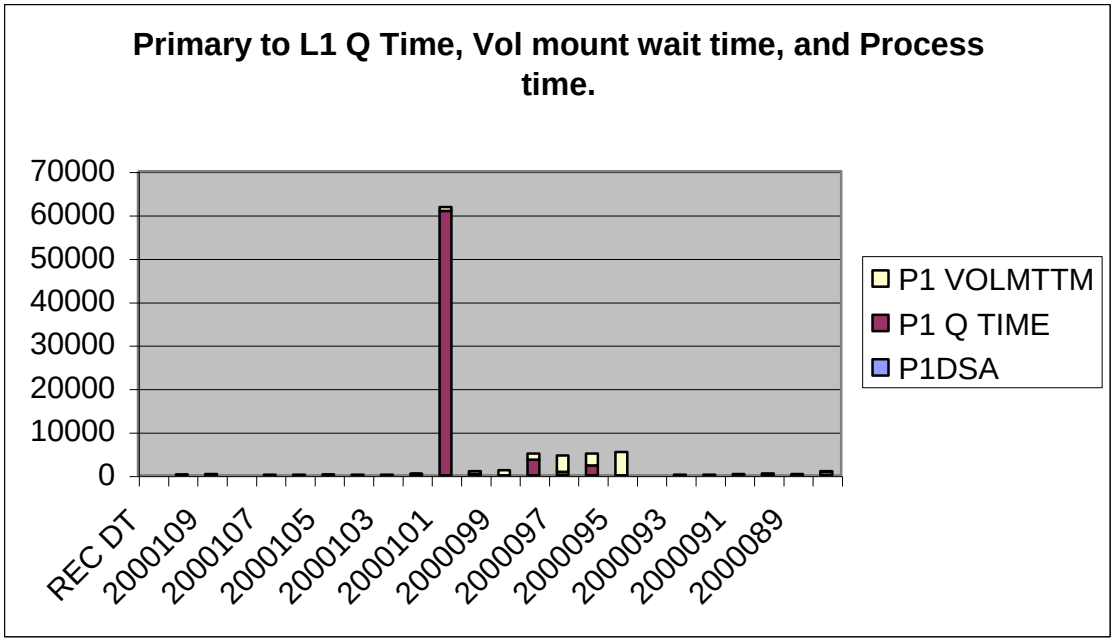


Figure 78: Health Report Chart - Primary to L1 Q Time, Vol Mount Wait Time, and Process Time

The above chart immediately shows you a long P1 Q Time. This could have been caused by a lost HSM tape, missed console message, or a list of other causes. For this example, because HSM waited so long for a request to be satisfied, it caused production to run past its scheduled window.

6 Summary

We hope you have benefited from this tour through HSM Reporter/Manager. There are many storage administration tasks that can be accomplished using the HSM Reporter/Manager panels. This tutorial did not cover all of the available options or features of HSM Reporter/Manager and is not intended to replace the *HSM Reporter/Manager User Guide* (document 002-0202) that came with the product. Its purpose it to provide you with enough information to get you started. The *HSM Reporter/Manager User Guide* will be your best source for information.

If you need additional information, or have questions or problems, contact Mainstar Technical Support at 1-425-458-0084, or click here for the Mainstar Technical Support web page.

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Appendix A

A Data Class Articles

This appendix contains copies of the Data Class Articles, Part 1 and Part 2, that were referenced in this tutorial.

A.1 The (forgotten?) Data Class (Part 1 of 2)

System Managed Storage (SMS) implementations void of data class assignments is missing out on many great features. Even if SMS has been implemented for quite some time, you can improve your storage environment by implementing data class assignments.

A.1.1 Eliminate Unknown DSORG

Data sets allocated with an unknown data set organization (DSORG) can not be migrated by HSM. This condition is caused by the DSORG being specified in the program but the program did not open the data set after allocation. To eliminate this situation, define a data class with no attributes (or a small amount of space attributes) and assign it to data sets with no valid data set organization. Assigning a data class causes the DSORG to default to physical sequential (PS) and allows HSM to migrate the data set.

The following are two examples of how to assign a data class called DCNOORG to data sets with no DSORG.

Example 1:

```
FILTLIST  &VALID_DSORG   INCLUDE('PS','PO','VS','DA')

WHEN (&DSORG NE '&VALID_DSORG')
       DO
          SET &DATACLAS = 'DCNOORG'
       EXIT
END
```

Example 2:

```
WHEN (&DSORG = '')
       DO
          SET &DATACLAS = 'DCNOORG'
       EXIT
END
```

A.1.2 Eliminate Pattern DSCBs

Since data class can be assigned to both SMS managed and non-managed data (including tape data sets), you can completely eliminate the need for a pattern DSCB when allocating generation data sets (GDS). If the implementation of pattern DSCBs did not include assignment of DCB parameters, you can simply test for data set type (&DSTYPE) in your ACS routines and assign a data class.

Example 1:

```
WHEN (&DSTYPE = 'GDS')
       DO
             SET &DATACLAS = 'DCGDG'
       EXIT
END
```

If the implementation of pattern DSCBs includes the assignment of DCB parameters, data classes with those DCB parameters need to be defined in ISMF. Use the DATACLAS JCL parameter on the DD statement to generate the proper DCB characteristics for the GDS.

A.1.3 VOLCOUNT

SMS 1.3 provided the VOLCOUNT parameter in the data class. This parameter allows for allocation across multiple volumes in the storage group, up to a maximum of 59. Indicating a VOLCOUNT greater than one in the data class allows data sets to be allocated multi-volume and reduces X-37 type ABENDs due to lack of space on one physical volume.

Be sure not to specify a VOLCOUNT greater than the number of volumes in the (subsequently) assigned storage group and do not assign a VOLCOUNT greater than one for partitioned data sets (PDS), SMS fails the allocation.

The following example shows how to check for a DSORG of PDS to avoid assigning a data class with a VOLCOUNT greater than one.

```
WHEN (&DSORG = 'PO')
        DO
             SET &DATACLAS = 'DCPDS'
        EXIT
OTHERWISE
        DO
             SET &DATACLAS = 'DCMULTIV'
        EXIT
END
```


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A.1.4 Other Space Constraint Relief Parameters

DFSMS 1.4 introduced new attributes in the data class to handle space constraints or X-37 type ABENDs. Use the space constraint relief attributes in addition to the VOLCOUNT parameter. If, during allocation, there isn't adequate space to satisfy the primary allocation, the Space Constraint Relief (Y) and Reduce Space Up To % (some value 0 – 99) attribute causes allocation to be re-tried. SMS attempts to either spread the space quantity over multiple volumes, allocating only a percentage (% value indicated in data class) of the requested quantity, or by using more than five extends to satisfy the allocation.

To use the space constraint relief features; indicate a value of Y (yes) in the Space Constraint Relief field. Indicate a value for the percentage that SMS should reduce the primary allocation by, for example 50, in the Reduce Space Up to % field and indicate a VOLCOUNT greater than one in the VOLCOUNT field. If you indicate zero in the Reduce Space Up to % field, SMS removes the 5-extent limit and retires the allocation across multiple volumes.

The number of extents used during initial allocation might result in fewer extents being subsequently available. For example, when the primary space allocation uses 10 extents then only 6 extents are left for allocation of the secondary quantity.

Compatibility PTFs are required for all DFSMS/MVS releases below 1.4 and for all DFP releases so that lower level systems can access all 255 extents of a VSAM component created on a DFSMS/MVS 1.4 system.

A.2 The (forgotten?) Data Class (Part 2 of 2)

A.2.1 System Managed Buffering

DFSMS 1.4 introduced System Managed Buffering for data sets in extended format. An extended format VSAM data set is similar to a PDSE in that they are “SMS only” type data sets.

System Managed Buffering is basically VSAM being guided on how to optimize buffers based on the Record_Access_Bias parameter specified in the data class or on the JCL AMP parameter. The manner in which buffers are processed refers to buffer management continuing as non shared resources (NSR), or being changed internally to local shared resources (LSR) to take advantage of LSR capabilities, which may include the use of hyperspace. The system selects buffering algorithms and buffer sizes based on the processing options that are specified when the data set is opened and on the data sets storage class attributes.

DFSMS 1.4 supports only KSDS VSAM (NSR) data sets. DFSMS 1.5 supports entry-sequenced data sets (ESDS), relative-record data sets (RRDS), variable-length relative-record data sets (VRRDS), and linear data sets (LDS). Extended format data sets other than KSDS can not be accessed on DFSMS/MVS releases prior to 1.5.

Once VSAM data sets are in extended format, they can take advantage of partial release, space allocation on candidate volumes and system managed buffering.

A.2.2 Partial Release

Space beyond the high-used control area can now be released to the system. You can specify partial release via the management class or in the JCL by specifying SPACE=(,,RLSE).

A.2.3 Space Allocation on Candidate Volumes

An extension to a new volume can use either the primary or secondary space allocation. Use the data class parameter Additional Volume Amount to indicate either the primary or secondary value.

A.2.4 System-Managed Buffering

All VSAM NSR data sets can take advantage of System Managed Buffering by specifying Record_Access_Bias SYSTEM in the data class or by using the new JCL keyword ACCBIAS=SYSTEM.

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A.2.5 Application Access Considerations

System Managed Buffering should be used for applications that access KSDS data sets using a standard access method such as COBOL II. Applications that use RRDS or VVRDS should access the data using relative record number and not relative byte address (RBA) for control interval access. Applications that use ESDS or LDS only require changes if the data is defined for extended addressability (EA). Extended addressability supports data sets larger than 4GB.

Extended Format data sets are support by HSM, DFDSS and ABARS during migration, backup, dump, ABACKUP and ARECOVER as long as the data sets are recovered to a SMS managed environment.

A.2.6 Restrictions

An alternate index can not be used with an extended format ESDS data set. KSDS data sets can not be defined with keyranges or with the index sequence set imbedded in the data (IMBED). Data sets defined for VSAM RLS can not use System Managed Buffering. VSAM RLS does not support linear data sets (LDS). Hyperbatch support is not available. HSM can not migrate or backup any VSAM extended format data set larger than 64K tracks.

A.2.7 Other Notable Facts

When performing IDCAMS REPRO processing on an extended format VSAM data set, be sure that the data set is not open. If the data set is open, and the data set extends, the updates may not get copied to the target catalog resulting in a discrepancy between the information in the VVDS and the new target BCS. The result is loss of information or the inability to access the data set.

Extended format support allows applications that use very large BDAM data sets to convert to VSAM RRDS and exploit sharing and data extensions.

A.2.8 Defining A Data Class for Extended Format Data Sets and System Managed Buffering

In the data class definition, set the Data Set Name Type to EXT, the IF EXT to either P (preferred) or R (required), and the Record_Access_Bias to S (system). Instruct applications to use the keyword DATACLAS and the data class name you assign in the REPRO or allocation JCL.

The following is an example of the new JCL keywords in support of System Managed Buffering.

```
ACCBIAS   record access bias
          USER           let the user control the buffering
          SYSTEM         invoke system-managed buffering
          SO             SMB with sequential optimization
          SW             SMB weighted for sequential processing
          DO             SMB with direct optimization
          DW             SMB weighted for direct processing
RMODE31   virtual storage location of buffers and control blocks
          BUFF           buffers above the 16MB line
          CB             control blocks above the 16MB line
          All            control blocks and buffers above the 16MB line
          NONE           control blocks and buffers below the 16MB line
          SMBDFR         record management PUT defer processing
          SMBHWT         LSR hiperspace weighting factor
          SMBVSP         Maximum data buffer space
```


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Reader's Comments

HSM Reporter/Manager Tutorial

We want to receive your comments about this manual. Your suggestions about the accuracy, clarity, and format of this document will be appreciated.

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Bellevue, WA 98009 USA

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Fax: 1-425-455-1992

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Fax: 1-425-455-1992

European Sales

Mainstar Software Corporation Ltd.
The Gables
Market Square
Princes Risborough
Bucks HP270AN
England
Tel: +44-(0)1844-347173
Fax: +44-(0)1844-346627

Asia Pacific Sales

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